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City University of Hong Kong



Pricing Kernel Puzzles

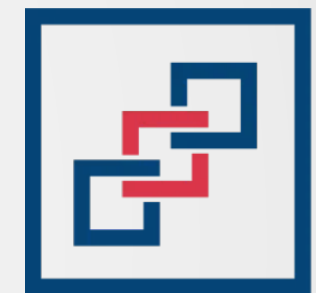
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All You Need is...



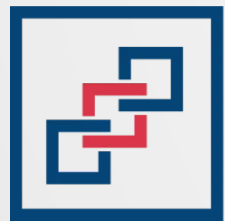
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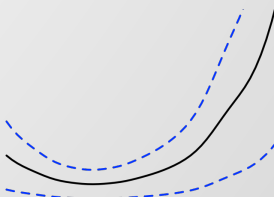
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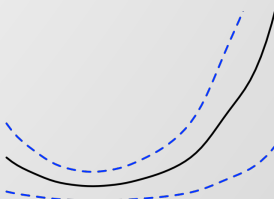


Two sides of the same Coin

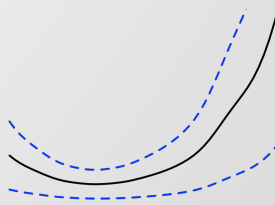
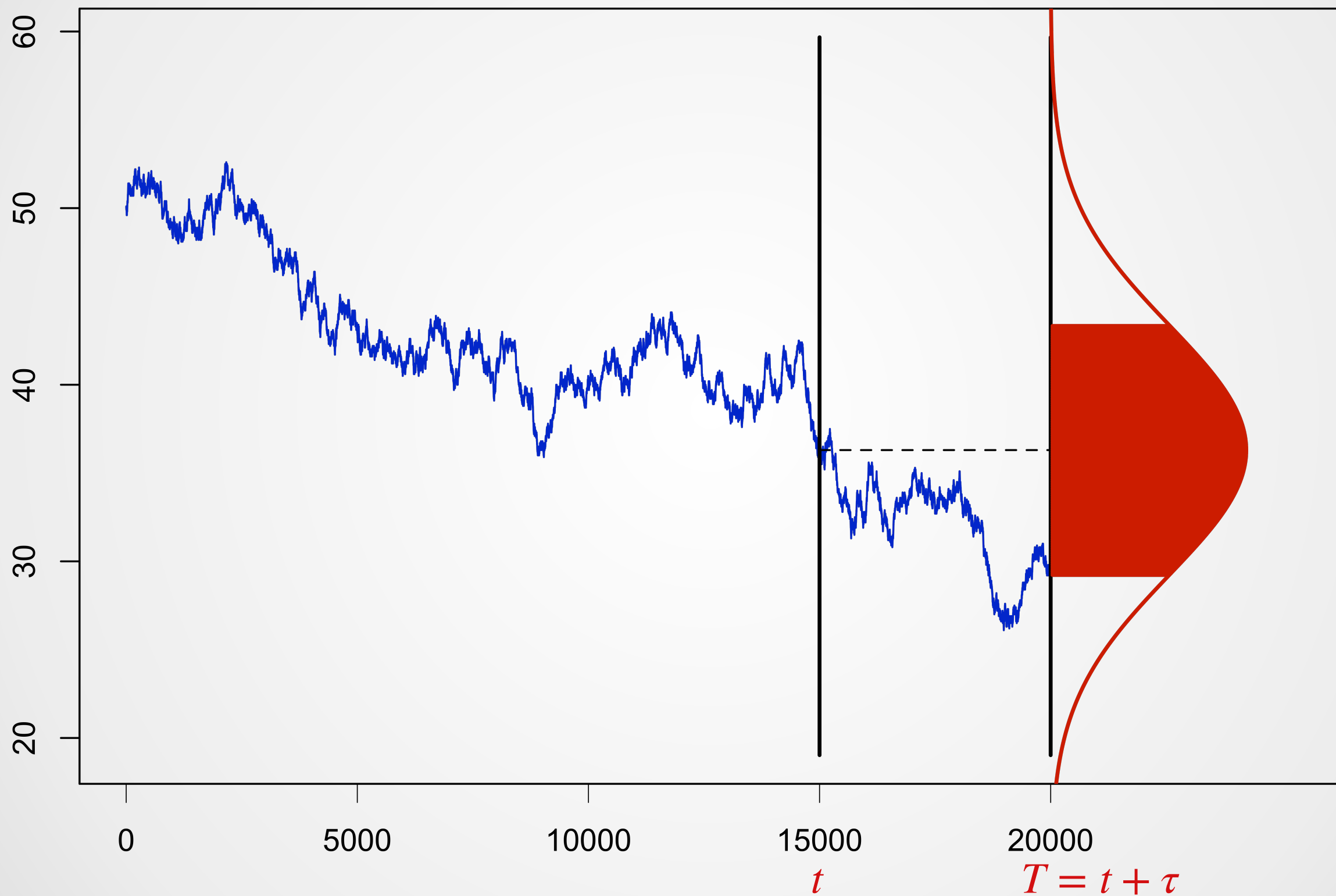
“Shiller and Fama’s works ultimately discuss the same findings, but interpreting them differently. What Shiller calls irrational bubbles and behavioural biases, Fama would call efficient markets and varying risk premia, **is nothing but the empirical observation that the stochastic discount factor varies a lot over time.**

Whether we label this as “irrational behavior” or “time-varying risk premia” is an hypothesis that is very hard to test empirically.”

MK Brunnermeier, Asset Pricing I, Princeton University



Stock, Prices, Options



Pricing Kernel

- Risk neutral valuation principle for pay-offs

$$p_t = \mathbb{E}_Q\{e^{-r\tau}\psi(S_T)\} = \int_0^\infty e^{-r\tau}\psi(x)q(x)dx = \int_0^\infty e^{-r\tau}\psi(x)\frac{q(x)}{p(x)}p(x)dx$$

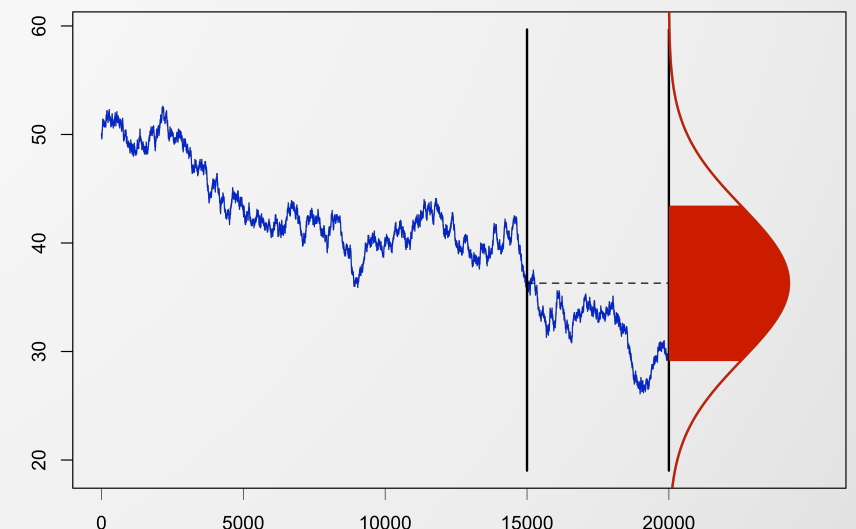
$K(x)$: Pricing kernel (PK)

$$= \int_0^\infty e^{-r\tau}\psi(x)K(x)p(x)dx = \mathbb{E}_P\{e^{-r\tau}K(S_T)\psi(S_T)\}$$

SDF: discounts payoffs across states

- ▶ $p(\cdot)$ physical density of stock S_T under measure P
- ▶ $q(\cdot)$ risk-neutral density of stock S_T under measure Q
- ▶ $\psi(S_T)$ pay-off, r interest rate, τ time to maturity
- Law of one price (Loop) and non-arbitrage

K exists and $K > 0$



PK Shape: Monotonically decreasing!

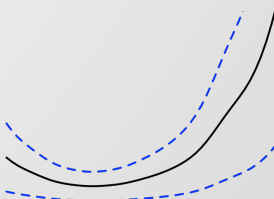
- ▣ Risk-free rate $r = 0$, payoff equals price $\psi(S_T) = S_T$
- ▣ Risk averse investors, assuming
 - ▶ Consumption C_T equals stock price $C_T = S_T$
 - ▶ Utility $U(C_T) = \sqrt{C_T}$, marginal utility $U'(C_T) = u(C_T) = 1/(2\sqrt{C_T})$
 - ▶ S_t based on utility: $S_t = E_P\{\beta u(S_T)S_T\}$
 - ▶ $K(S_T) = \beta u(S_T)$
- ▣ Calculate $K(S_T)$ based on
 - ▶ Constraints: $\int_0^\infty q(x)dx = 1 \succ \int_0^\infty K(x)P(x)dx = \int_0^\infty \beta u(C_t)P(x)dx = 1$

Given



[EPK and Utility](#)

State	$P(S_T)$	S_T	$u(C_T)$	$K(S_T)$	$q(S_T)$	S_0
Bad	0.5	80	0.056	1.1	0.55	98
Good	0.5	120	0.046	0.9	0.45	



PKs in practise

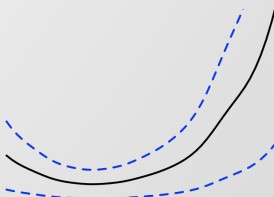
- ▣ RW goes long $S_t = 41.85$ with $\tau = 1M$
- ▣ RW discloses $P(S_T = 45) = 0.9$, $P(S_T = 40) = 0.1$, $r = 0.05$
- ▣ What is RW's PK $K(S_T)$, or Prefs?

$$\text{Stock Price: } 41.85 = 0.9 \times 45 \times K_u + 0.1 \times 40 \times K_d$$

$$\text{Bond Price: } e^{-0.05 \times 1/12} = \underbrace{0.9 \times K_u}_{\psi_u} + \underbrace{0.1 \times K_d}_{\psi_d}$$

Arrow-Debreu price:

$$\text{Q density: } q(S_T) = P(S_T)K(S_T)$$

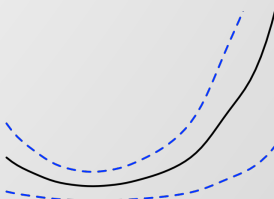


PKs in practise

Given



State	$P(S_T)$	S_T	S_t	$K(S_T)$	$\psi(S_T)$	$Q(S_T)$
Up	0.9	45	41.85	0.448	0.403	0.405
Down	0.1	40		5.927	0.593	0.595



Pricing Puzzle



- ▣ „Risk aversion“ enforces $\hat{K}$ to be monotone & non-increasing
- ▣ Empirically, $\hat{K}$ is often non monotone
- ▣ Against economic principles:
 - ▶ Local „humps“ reveal preference for higher risk (or lower returns) in some return regions
 - ▶ Additional factors yield large risk premium (Chabi-Yo et al, 2007)
 - ▶ Investors without risk-averse preference (Ziegler, 2007)
 - ▶ Biased beliefs (Bakshi et al, 2010)
 - ▶ Flaws in existing estimators (Linn et al. 2018, RFS)



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JOURNAL ARTICLE

Pricing Kernel Monotonicity and Conditional Information

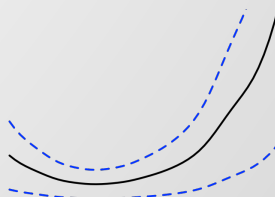
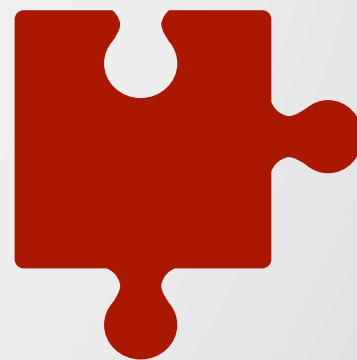
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Matthew Linn, Sophie Shive, Tyler Shumway ✉

The Review of Financial Studies, Volume 31, Issue 2, February 2018, Pages 493–531,

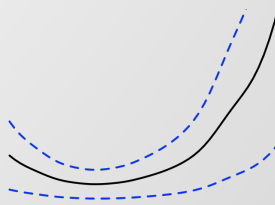
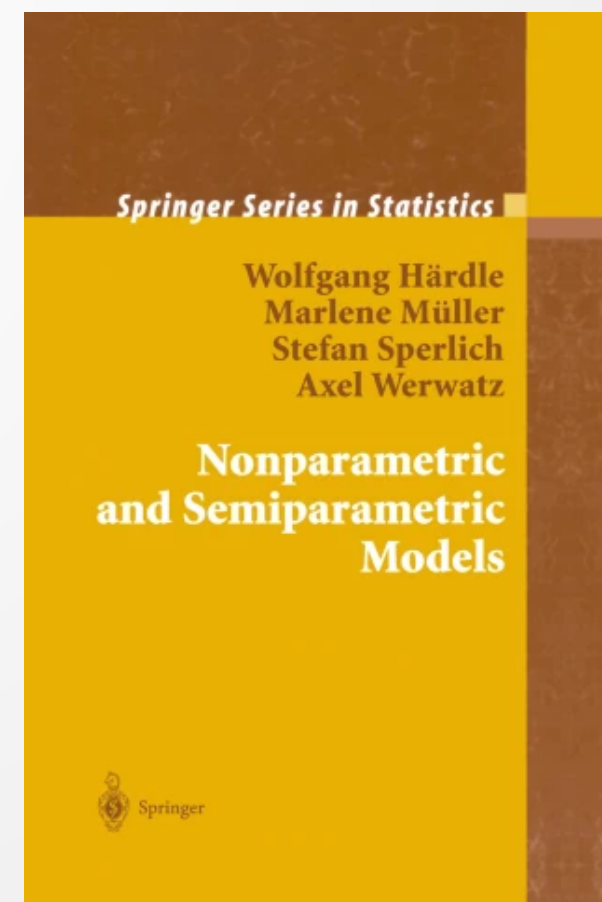
<https://doi.org/10.1093/rfs/hhx095>

Published: 21 August 2017



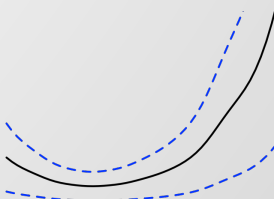
Find a scenario for testing volatility & risk preference

- Bitcoin market is different
 - ▶ Volatility
 - ▶ Expected utility (Grith et al., 2017)
- Pricing kernel (PK) puzzle in Bitcoin (BTC) option markets
 - ▶ Classical estimation is not monotonically decreasing
 - ▶ Will the monotonicity claim by Linn et al (2018) still hold?
- PK estimation for BTC options
 - ▶ Bandwidth h , canonical kernel
 - ▶ Moments and knots



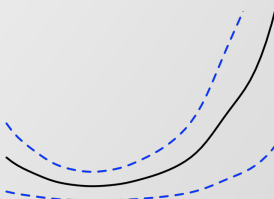
Pricing kernel for BTC and SP500 options

- ▣ Collect BTC and SP500 option data
- ▣ Estimate the PK using classical method (Rookley, 1997)
 - ▶ Estimate Q density and P density separately
 - ▶ Pricing kernel calculated using Q density and P density
- ▣ Estimate PK using CDI method
 - ▶ Estimate Q density, same as the classical method
 - ▶ Do not need to estimate the P density
- ▣ Compare **classical PK results** with **CDI method**
- ▣ Confidence bands

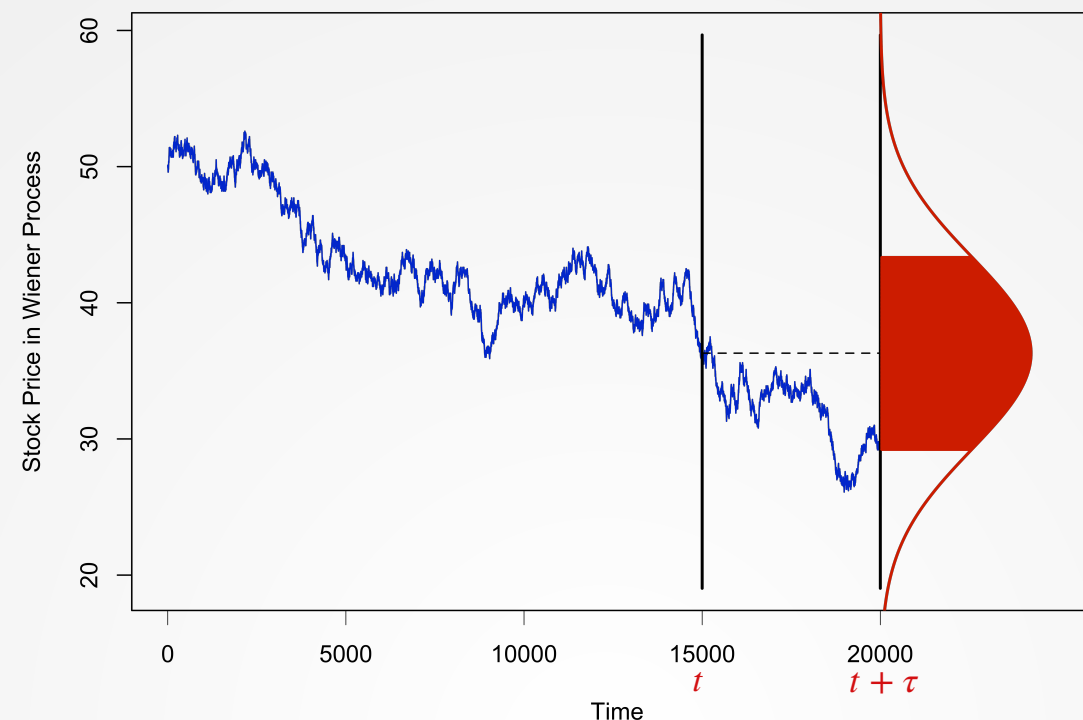


Outline

1. Motivation ✓
2. Pricing Kernel and Risk Preferences
3. Classical PK Estimation
4. CDI approach
5. PK of BTC options
6. Classics vs CDI
7. Volatility jumps
8. B-splines and kernels
9. References



Asset prices



SFEwienerdens 

τ : time to maturity

r : interest rate

μ : stock drift

σ : volatility

Wiener process

$$\{W_t; t \geq 0\}$$

$$E[W_t] = 0, \text{Var}[W_t] = t$$

$$\text{Cov}(W_{t+\tau}, W_t) = t$$

$$(W_{t+\tau} - W_t) \sim \mathcal{N}(0, \tau)$$

Increment $dW_t \sim \mathcal{N}(0, dt)$

Real world in P -GBM:

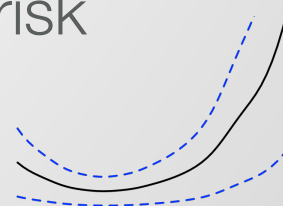
$$S_t = S_0 \exp \left\{ \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t^P \right\}$$

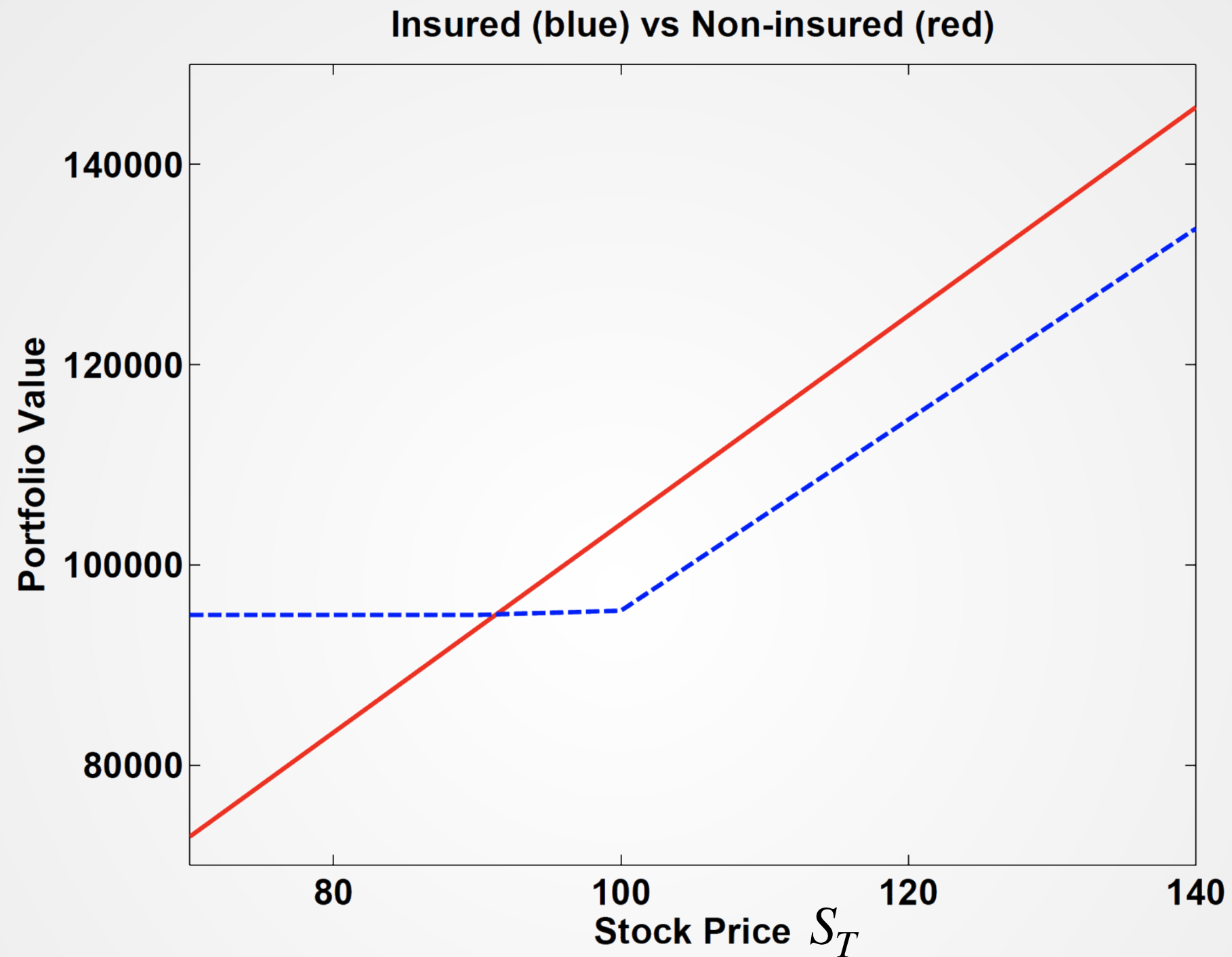
Risk-neutral world Q -GBM

$$S_t = S_0 \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) t + \sigma W_t^Q \right\}$$

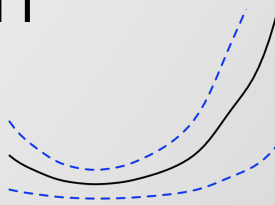
$$W_t^Q = W_t^P + \gamma t$$

$$\gamma = \frac{\mu - r}{\sigma} \text{ the market price of risk}$$





Portfolio value: not insured, insured $\psi(S_T)$



Theoretical PK in the Black-Scholes world

- State Price Density (SPD):

$$q(S_T) = e^{r\tau} \frac{\partial^2 C}{\partial X^2} \Big|_{X=S_T}$$

- Black Scholes framework, European call options

$$C(S_t, X, \tau, r, \sigma^2) = S_t \Phi(d_1) - X e^{r\tau} \Phi(d_2)$$

d_1 and d_2 are known functions of σ^2 , τ , X , and S_T

- Fischer Black-Myron Scholes: $S_T = S_t \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) \tau + \sigma (W_T^Q - W_t^Q) \right\}$

lognormal distributions:

$$q(S_T) = \frac{1}{S_T \sqrt{2\pi\sigma^2\tau}} \exp \left[-\frac{\{ \log(S_T/S_t) - (r - \sigma^2/2)\tau \}^2}{2\sigma^2\tau} \right]$$

C : call option price

X : strike price

σ : volatility

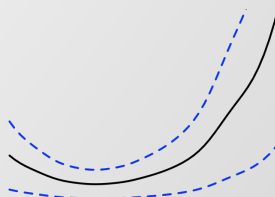
S_t : stock price at time t

T : maturity

τ : time to maturity

r : interest rate

μ : stock drift



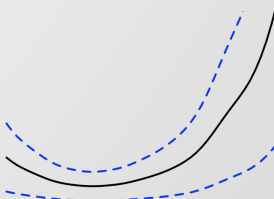
Investor preference analysis

Real world under physical probability: $S_t = S_0 \exp \left\{ \left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t^P \right\}$

$$S_T = S_t \exp \left\{ \left(\mu - \frac{1}{2}\sigma^2 \right) \tau + \sigma(W_T^P - W_t^P) \right\}$$

$$p(S_T) = \frac{1}{S_T \sqrt{2\pi\sigma^2\tau}} \exp \left[-\frac{\{\log(S_T/S_t) - (\mu - \sigma^2/2)\tau\}^2}{2\sigma^2\tau} \right]$$

- ▣ Risk-free rate r
- ▣ Volatility σ , Drift μ (under P)
- ▣ P -Brownian motion W_t^P



Investor preference analysis

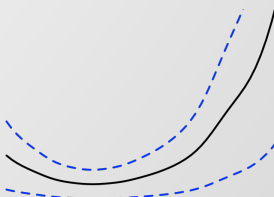
▣ Pricing kernel

$$K(S_T) = \frac{q(S_T)}{p(S_T)}$$

BS model consistent with log-normal distribution of S_t .

$$q(S_T) = \frac{1}{S_T \sqrt{2\pi\sigma^2\tau}} \exp\left[-\frac{\{\log(S_T/S_t) - (r - \sigma^2/2)\tau\}^2}{2\sigma^2\tau}\right]$$

$$p(S_T) = \frac{1}{S_T \sqrt{2\pi\sigma^2\tau}} \exp\left[-\frac{\{\log(S_T/S_t) - (\mu - \sigma^2/2)\tau\}^2}{2\sigma^2\tau}\right]$$



Investor preference analysis

$$K(S_T) = \frac{q(S_T)}{p(S_T)} = \exp \left\{ -\frac{1}{2} \frac{2 \log(S_T/S_t)(\mu - r) + \left(r - \frac{1}{2}\sigma^2\right)^2 \tau - \left(\mu - \frac{1}{2}\sigma^2\right)^2 \tau}{\sigma^2} \right\}$$

$$= \left(\frac{S_T}{S_t}\right)^{-\frac{\mu - r}{\sigma^2}} \exp \left\{ \frac{(\mu - r)(\mu - \sigma^2 + r)\tau}{2\sigma^2} \right\} = a \left(\frac{S_T}{S_t}\right)^{-b}$$

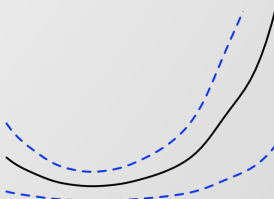
Where $a = \exp \left\{ \frac{(\mu - r)(\mu + r - \sigma^2)\tau}{2\sigma^2} \right\}$ and

$b = \frac{\mu - r}{\sigma^2} \geq 0$ constant coefficient of relative risk aversion of a power utility function

If $\mu - r > 0$ then $\gamma > 0$ and we look at a risk-averse representative investor

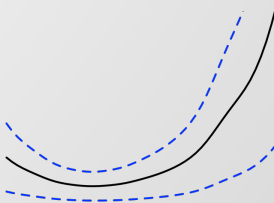
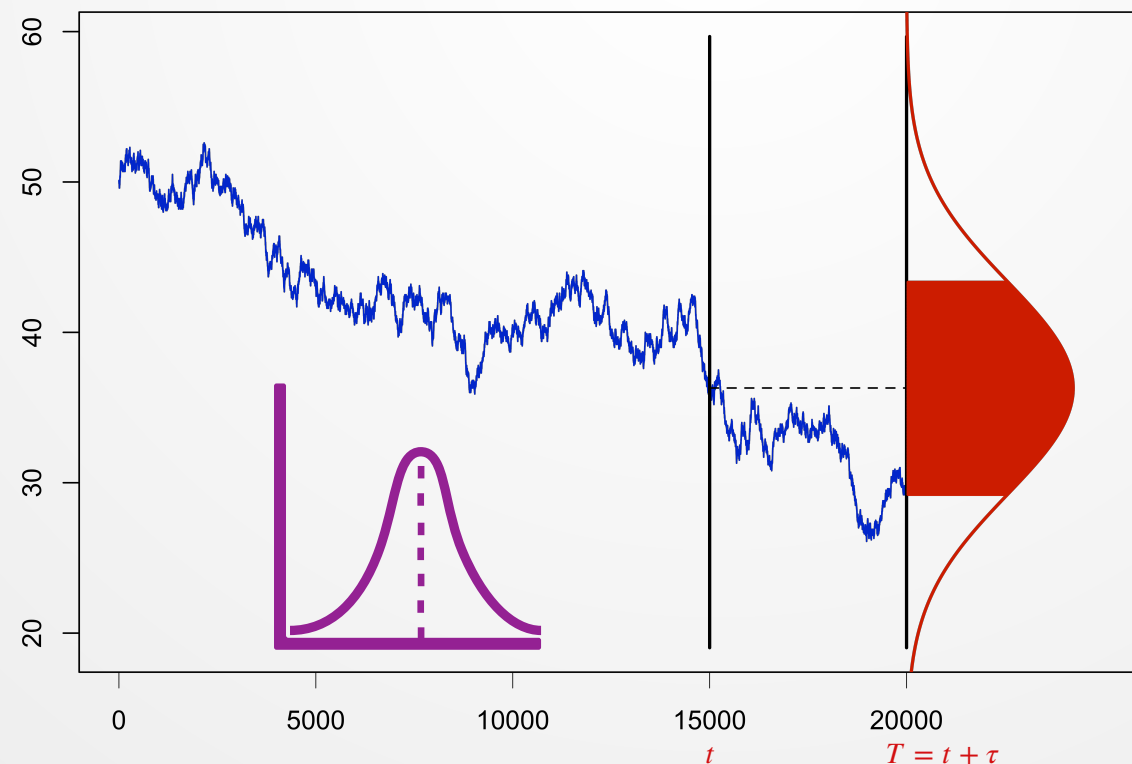


[EPK and Utility](#)



Literature

- ▣ Breeden & Litzenberger (1978): Risk neutral density (RND)
 - ▶ Second derivative of call option price wrt strike X
- ▣ Ait-Sahalia & Lo (2000); Jackwerth (2000)
 - ▶ Classical nonparametric Stochastic Discounted Factor (SDF) estimator: $\text{RND}/(\text{physical density based on historical returns})$
 - ▶ Decreasing function of market return

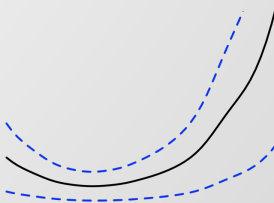


Econometric Approach

- ▣ Separate estimation of risk-neutral density and physical density

$$\widehat{K}(S_T) = \frac{\widehat{q}(S_T)}{\widehat{p}(S_T)}$$

- ▶ $\widehat{q}$ estimated by option prices ($\partial^2 C / \partial X^2$)
- ▶ $\widehat{p}$ estimated non parametrically using historical returns



Classical Method

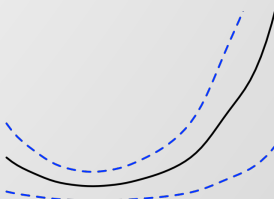
- ▣ $dF^P(x_{t+\tau}|\mathcal{F}_t)$ is physical density and $dF^Q(x_{t+\tau}|\mathcal{F}_t)$ is risk-neutral density and x_t is return at time point t
- ▣ SDF=PK of Jackwerth (2000) and Ait-Sahalia and Lo (2000)

$$m_t(x_{t+\tau}) = e^{-rs} \frac{dF^Q(x_{t+\tau}|\mathcal{F}_t)}{dF^P(x_{t+\tau}|\mathcal{F}_t)} \quad x_{t+\tau}: \log(S_T/S_t)$$

- ▣ Estimate risk-neutral and physical densities separately

$$m_t(x_{t+\tau}) = e^{-r\tau} \underbrace{\frac{dF^Q(x_{t+\tau}|\mathcal{F}_t)}{dx}}_{q(x_{t+\tau})} \bigg/ \underbrace{\frac{dF^P(x_{t+\tau}|\mathcal{F}_t)}{dx}}_{p(x_{t+\tau})}$$

- ▣ Strong assumption: information set for Q and P density is the same
- ▣ Can this assumption hold in volatile markets?



Estimation Process

Y_i : option price

x_i : strike price

$C()$: Call price function

ε_i : heterogeneity factors

- ▣ $q(\cdot)$ estimated nonparametrically

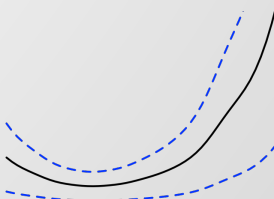
$$Y_i = C(x_i) + \sigma(x_i)\varepsilon_i, \quad i = 1, \dots, n$$

- ▣ $C(\cdot)$ approximated by local polynomial

$$C(u, x) = \sum_{j=0}^d C_j(x)(x - u)^j$$
$$C_j(x) = C^{(j)}(x)/j!, \quad j = 0, \dots, d$$

- ▣ Local Gaussian quasi likelihood

$$L\{Y, C(x)\} = -\{Y - C(x)\}^2 / \{2\sigma^2(x)\}$$



Estimation Process

- Function $C(\cdot)$ can be approximated by:

$$\hat{\mathcal{C}}(x) = \arg \max_{\mathcal{C}(x)} n^{-1} \sum_{i=1}^n K_h(x_i - x) L\{Y_i, C(x, x_i)\}$$

Kernel function

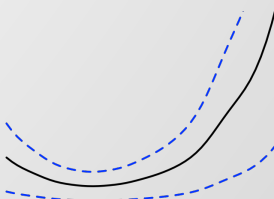
$$\mathcal{C}(x) = \{C_0(x), C_1(x), \dots, C_d(x)\}^\top$$

$$K_h(u) = K(u/h)/h$$

- Optimisation ($d = 3$):

$$A(X) \stackrel{\text{def}}{=} n^{-1} \sum_{i=1}^n K_h(x_i - x) \frac{\partial L\{Y_i, C(x, x_i)\}}{\partial C} \mathcal{X}_i = 0$$

$$\mathcal{X}_i \stackrel{\text{def}}{=} \{1, x_i - x, (x_i - x)^2, (x_i - x)^3\}^\top$$



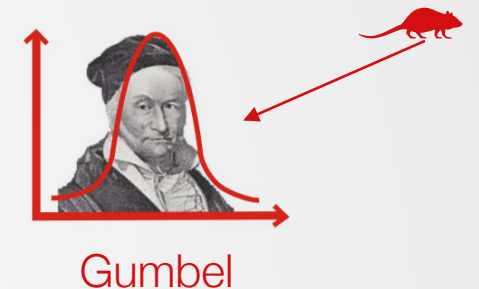
Estimation Process

- ▣ The physical density $p(x)$ is approximated as follows:

$S_t, \dots, S_{t+n+\tau-1}, (t+n+\tau-1 < T):$

$$\hat{p}(x) = n^{-1} \sum_{j=0}^{n-1} K_h\{x - \log(S_{t+j+\tau}/S_{t+j})\}$$

Kernel function



- ▣ The density of log-returns can be estimated as:

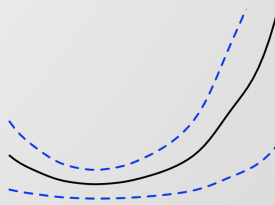
$$\hat{p}(x) = S_t \exp(x) \hat{p}\{S_t \exp(x)\}$$

$$\sup_{x \in E} |\hat{p}(x) - p(x)| = \mathcal{O}\{(nh_n/\log n)^{-1/2} + h_n^2\}$$

- ▣ EPK $\hat{K}(x) = \hat{q}(x)/\hat{p}(x)$

$$\sup_{x \in E} |\hat{K}(x) - K(x)| = \sup_{x \in E} \left| \frac{\hat{q}(x) - q(x)}{p(x)} - \frac{\hat{p}(x) - p(x)}{p(x)} \frac{q(x)}{p(x)} - \frac{\{\hat{q}(x) - q(x)\} \{\hat{p}(x) - p(x)\}}{p^2(x)} \right|$$

$$+ \mathcal{O}[\max\{(nh_n^{-1/2}) + h_n^2, h_n^{-2}\{nh_n/\log n\}^{-1/2} + h_n^2\}]$$



Conditional density integration (CDI) method

- Since $F(X) \sim U(0,1)$, it follows that

$$\int_{-\infty}^{X_{t+\tau}} dF^P(x_{t+\tau}|\mathcal{F}_t) = \int_{-\infty}^{X_{t+\tau}} \frac{dF^P(x_{t+\tau}|\mathcal{F}_t)}{dF^Q(x_{t+\tau}|\mathcal{F}_t)} dF^Q(x_{t+\tau}|\mathcal{F}_t)$$

Left hand: $F_t^P(X_{t+\tau})$

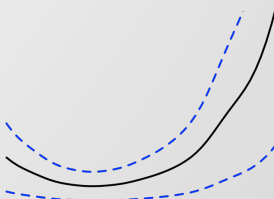
$$= \int_{-\infty}^{X_{t+\tau}} \left\{ \frac{dF^Q(x_{t+\tau}|\mathcal{F}_t)}{dF^P(x_{t+\tau}|\mathcal{F}_t)} \right\}^{-1} dF^Q(x_{t+\tau}|\mathcal{F}_t)$$

- Uniqueness given by

Recall: $PK = dQ/dP$

$$\int_{-\infty}^{X_{t+\tau}} g_{t,\tau}(y) dF^Q(y|\mathcal{F}_t) \sim U(0,1)$$

View the $F_t^P(X_{t+\tau})$ as a function of $m(X_{t+\tau})$, where $m(u) = \int_{-\infty}^u g_{t,\tau}(v) q_t(v) dv$.



CDI & nonparametric regression

$$Y = m(x) + \varepsilon$$

$$Y_i - m(X_i) = \varepsilon_i \sim U(0,1)$$

Observe

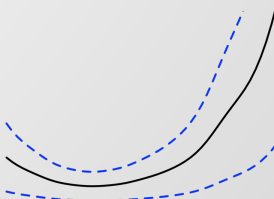
$$Y_i = m(X_{i+\tau}) + \varepsilon_i = \int_{-\infty}^{X_{i+\tau}} g_{\tau}(v) q(v) dv + \varepsilon_i$$

t dropped!

$$Y_i = \int_{-\infty}^{X_{i+\tau}} g_{\tau}(v) \hat{q}(v) dv$$

$= q(v) + \eta_i$

- Employ B-splines to approximate $g_{\tau}(v)$?



Estimation I

- Assume cubic B-spline

$$g(y) = \sum_{j=1}^J \theta_j B_j(y)$$

- Hence,

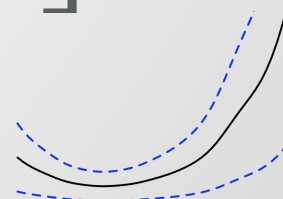
$$\int_{-\infty}^{X_{t+\tau}} g(y) dF^Q(y | \mathcal{F}_t) = \sum_{j=1}^J \theta_j \int_{-\infty}^{X_{t+\tau}} B_j(y) dF^Q(y | \mathcal{F}_t)$$



- Use GMM to get $\hat{\theta}$

$$\hat{\theta} = \operatorname{argmin}_{\theta \in \mathbb{R}^J} \sum_{i=1}^m \left[\frac{1}{T} \sum_{t=1}^T \left\{ \sum_{j=1}^J \theta_j \int_{-\infty}^{X_{t+\tau}} B_j(y) dF^Q(y | \mathcal{F}_t) \right\}^i - \frac{1}{i+1} \right]^2$$

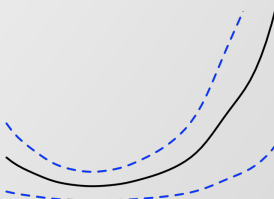
moments of the uniform distribution



Estimation II

- Sensitive to moments m and basis functions J
 - Plug-in estimate given by

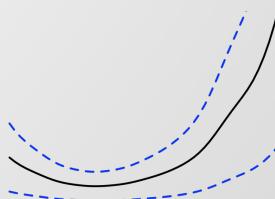
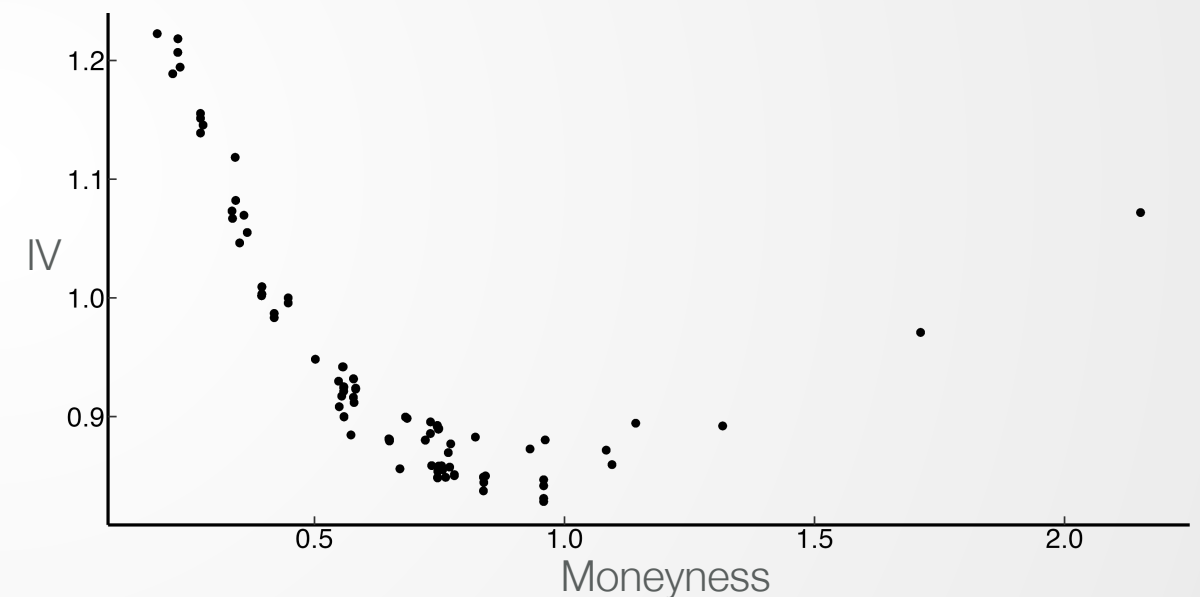
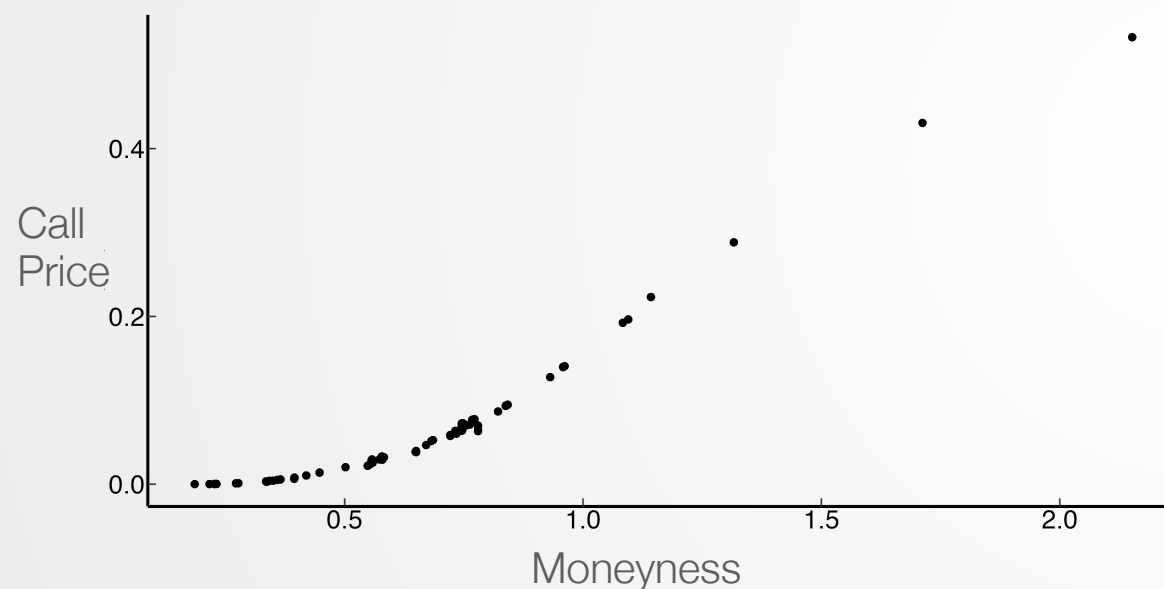
$$\hat{g}(u) = \sum_{j=1}^J \hat{\theta}_j B_j(u)$$
 - EPK follows as $\hat{K}(u) = \{\hat{g}(u)\}^{-1}$
 - ▶ All moments m are equally weighted
 - $\hat{q}$ is based on tail distribution assumptions
 - ▶ Figlewski (2009): GEV brings not smoothly tail
 - ▶ Linn et al. (2018): GPD change true $\hat{q}$ in the middle portion
 - ▶ Jiao et al.(2021): left tail to the GPD but not modify middle portion
 - ▶ Zhang & Dai (2025): CV Model to improve $\hat{q}$ for short-term maturity



So let us do it!

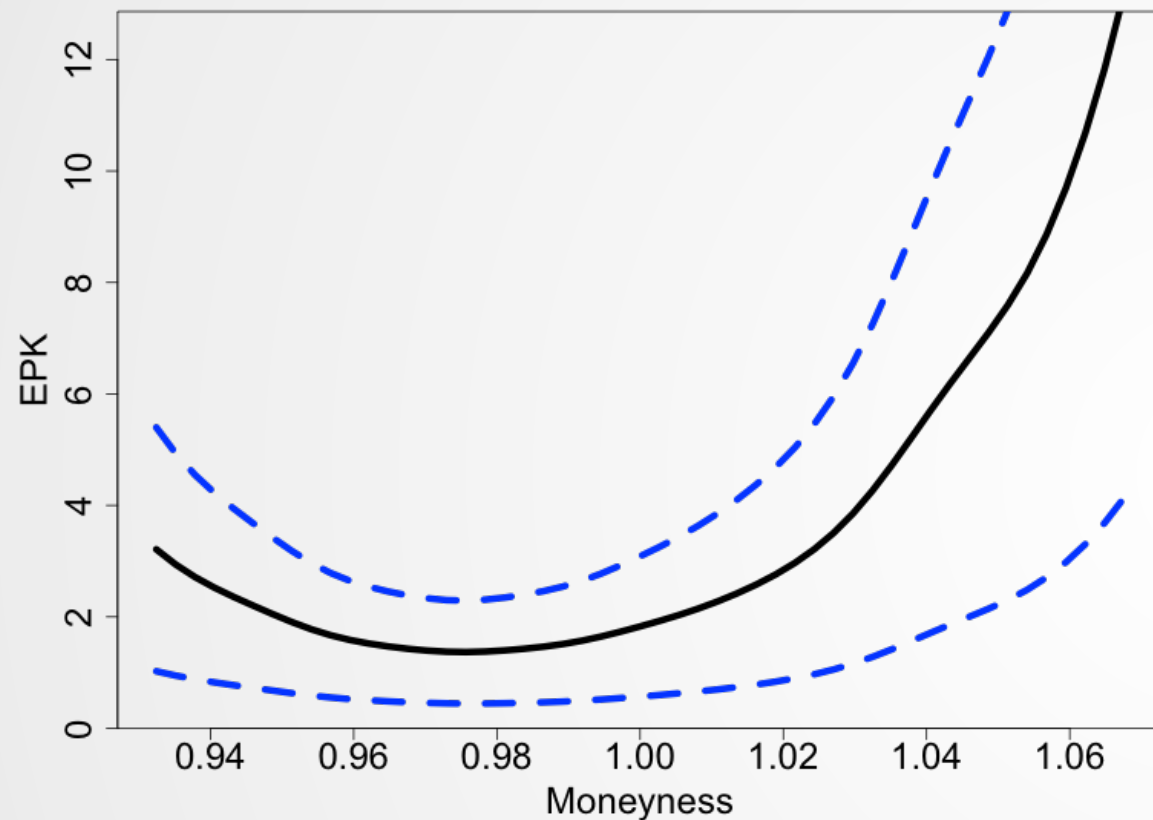


- BTC price 20200801 = 11789.63
- Call options: 53
- Display the IV and Call scatterplots on moneyness and τ

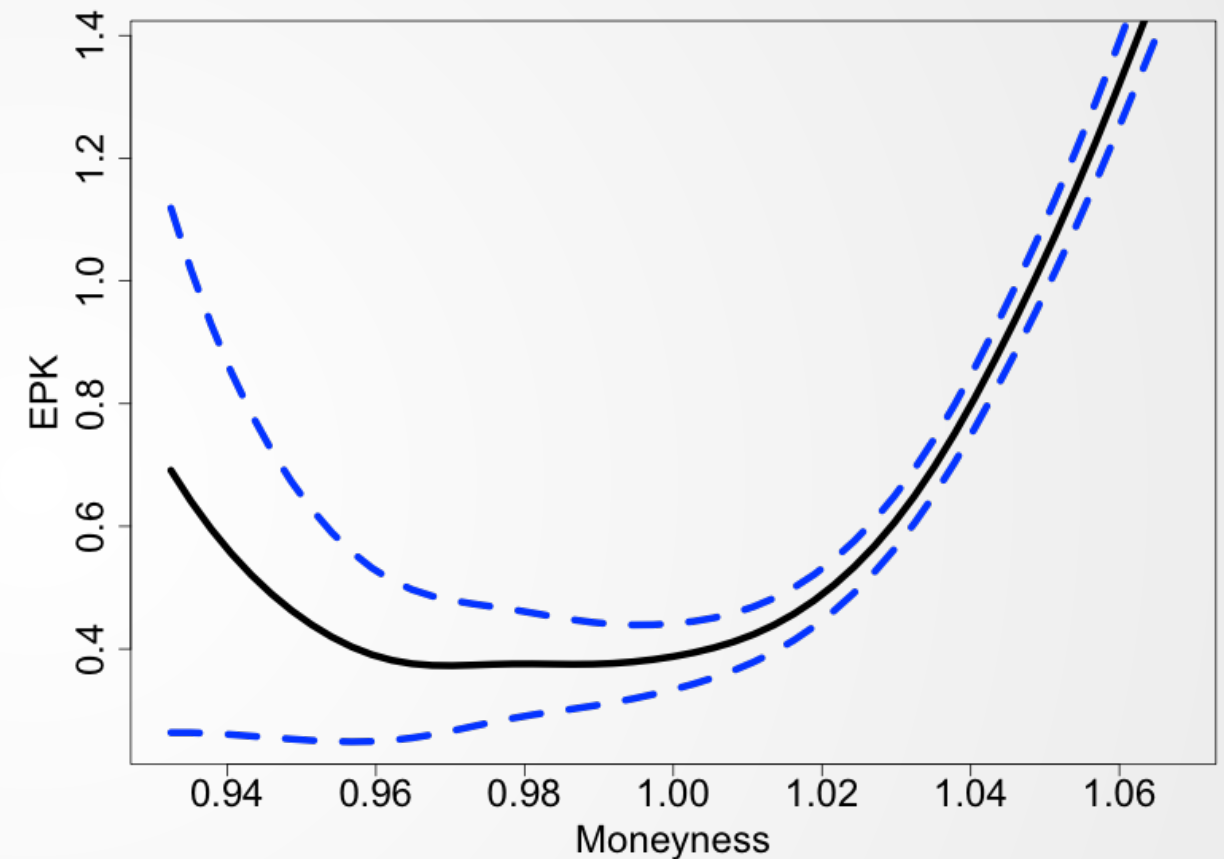


„Pricing Puzzle“ using classical methods

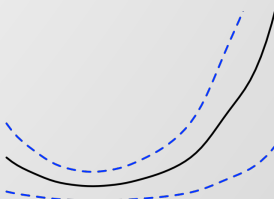
- Non monotonicity is discovered 



Classical result of S&P 500 Index Option
2017-01-27 $\tau = 32$
bandwidth: 0.08, confidential level: 95%

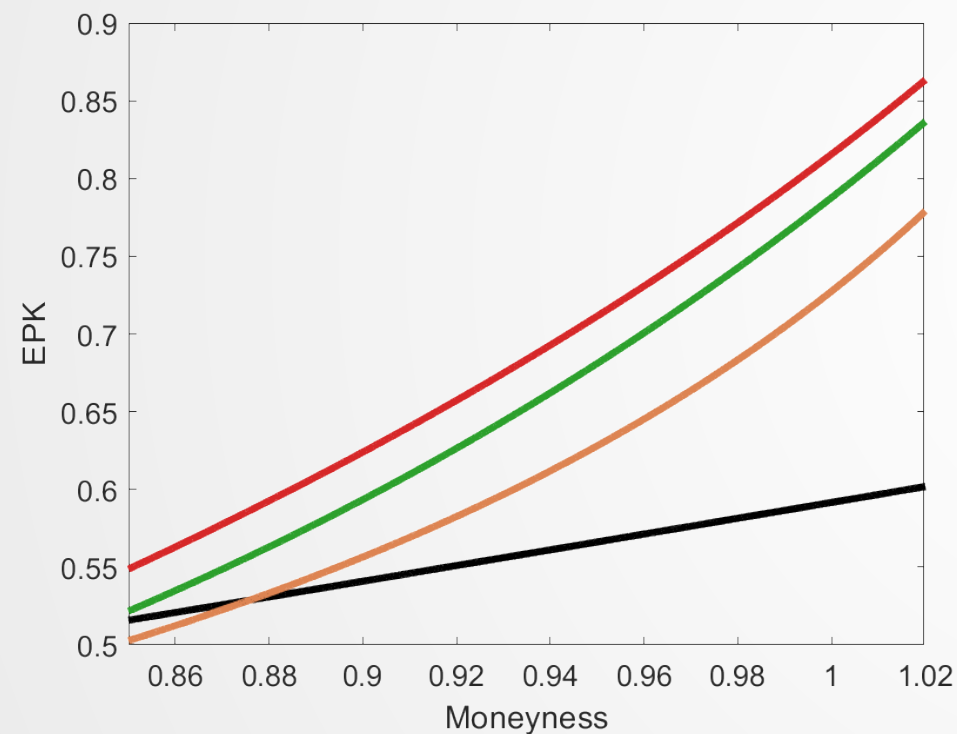


Classical result of S&P 500 Index Option
2018-12-12 $\tau = 26$
bandwidth: 0.08, confidential level: 95%



Pricing kernel using CDI method

- 2017-01-27 $\tau = 32$, increase monotonically
- 2018-12-12 $\tau = 26$, monotonicity result is sensitive to the choices of moments and knots

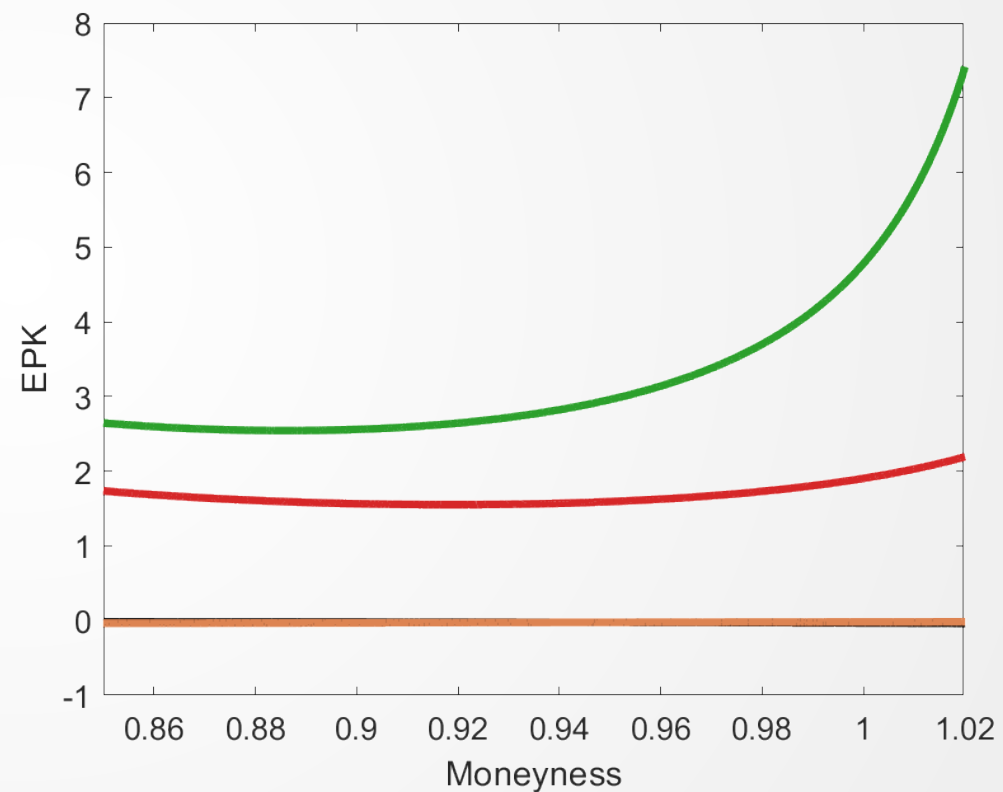


CDI result of S&P 500 Index Option

2017-01-27 $\tau = 32$

4 moments, 4 knots; 5 moments, 5 knots;

6 moments, 6 knots; 7 moments, 7 knots

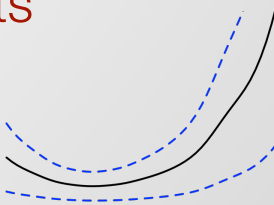


EPK of S&P 500 Index Option

2018-12-12 $\tau = 26$

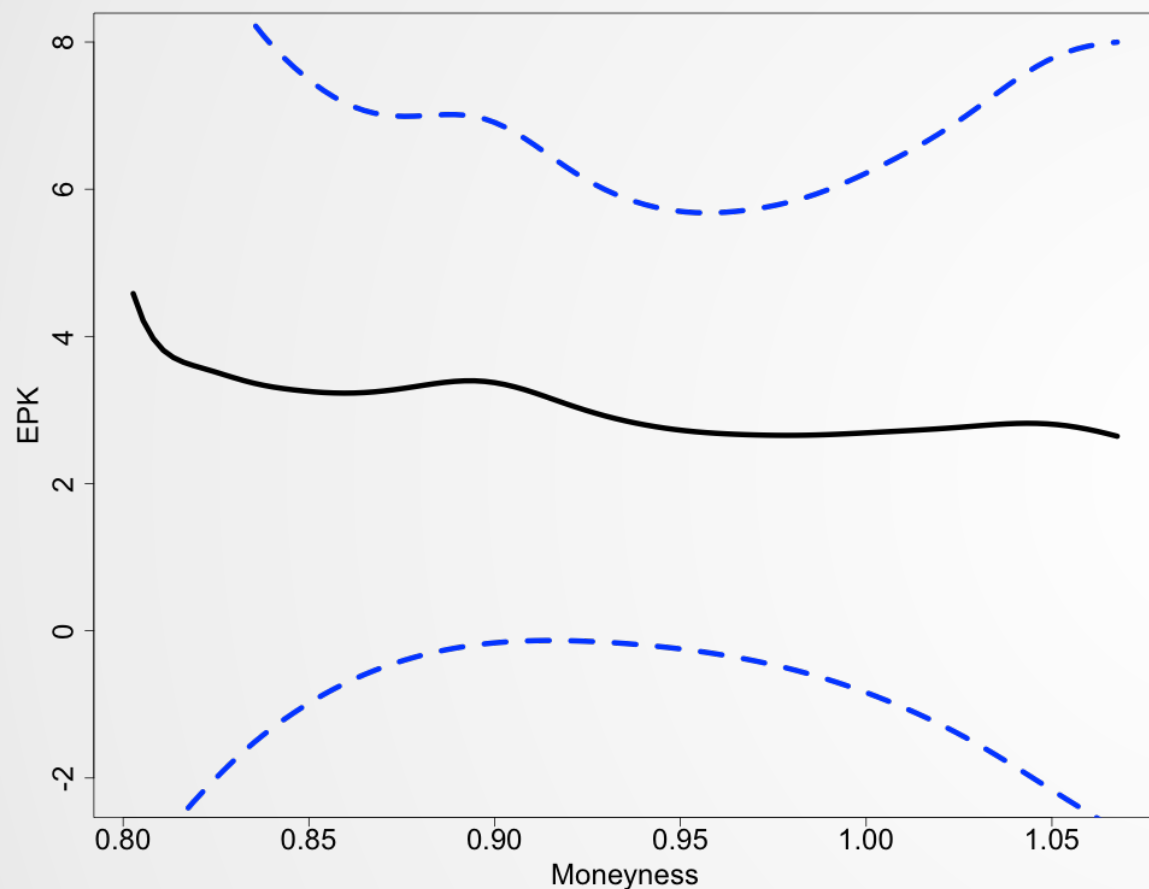
4 moments, 4 knots; 5 moments, 5 knots;

6 moments, 6 knots; 7 moments, 7 knots



Pricing kernel- Bitcoin option data

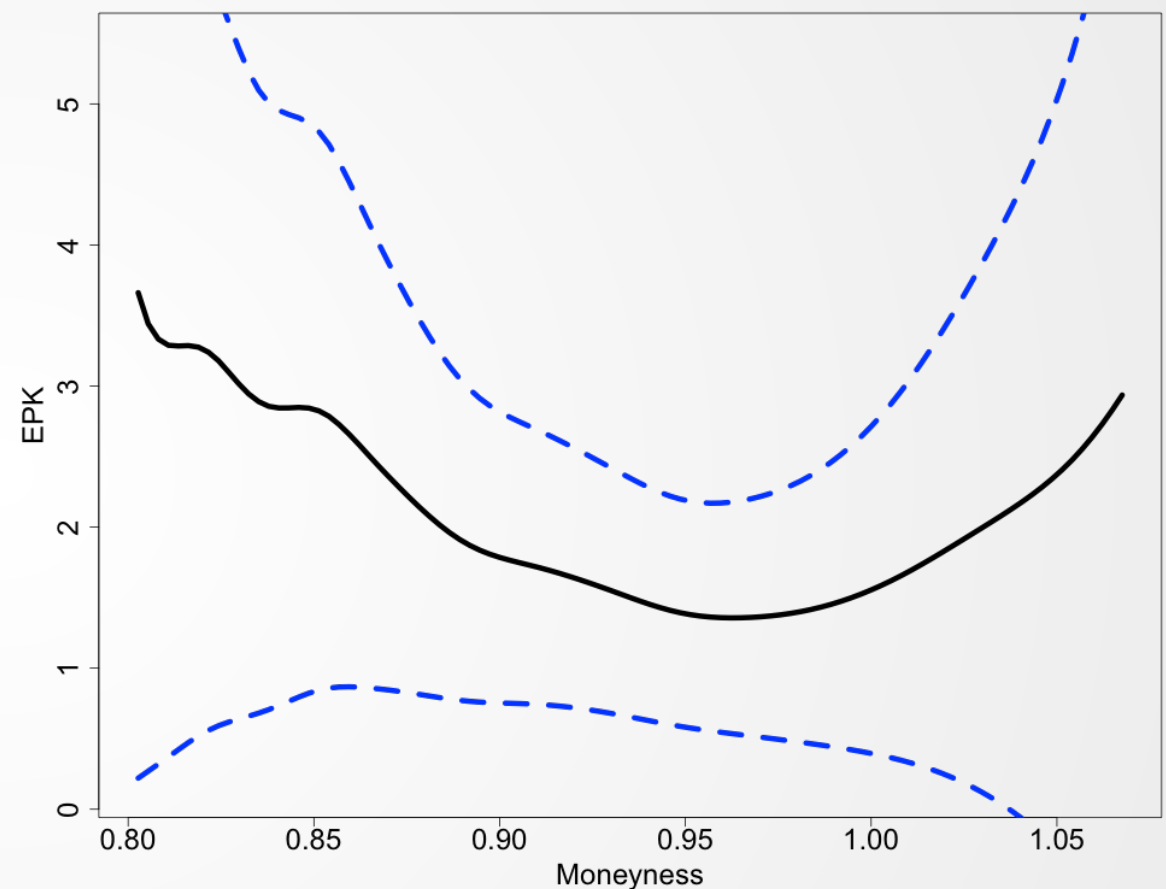
Classical EPK in Bitcoin option market



Classical result of Bitcoin Option

2022-11-17 $\tau = 8$

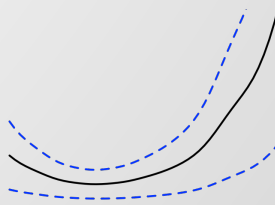
bandwidth: 0.08, confidential level: 95%



Classical result of Bitcoin Option

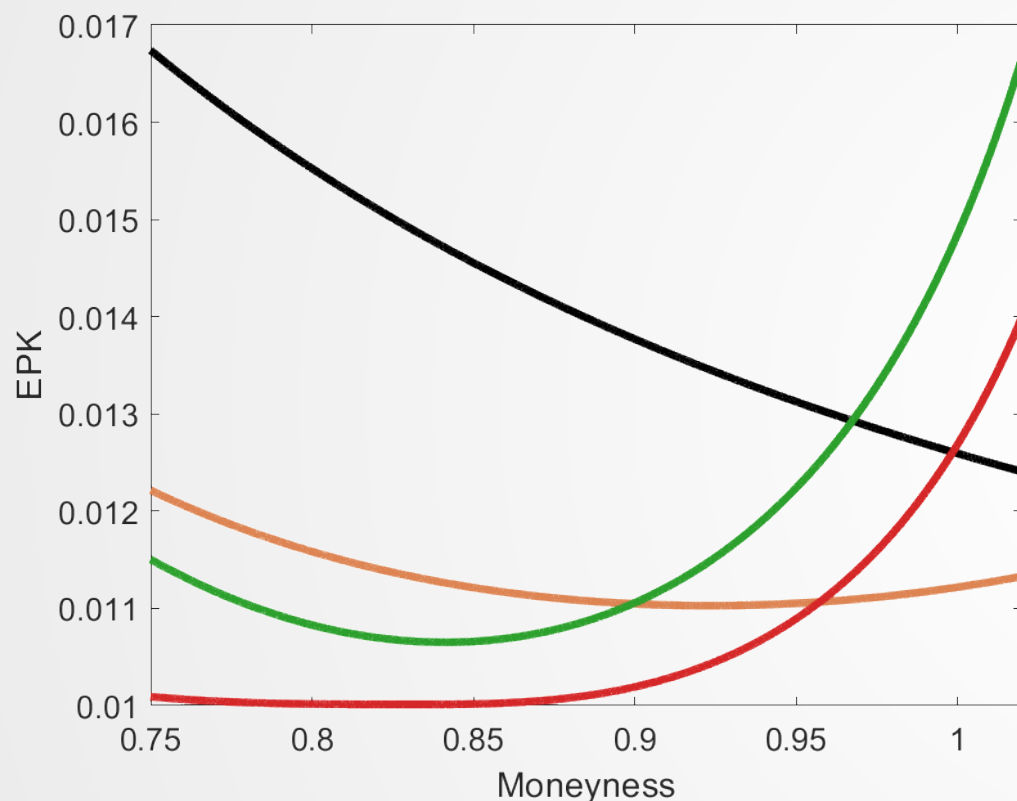
2022-10-29 $\tau = 13$

bandwidth: 0.08, confidential level: 95%



Pricing kernel- Bitcoin option data

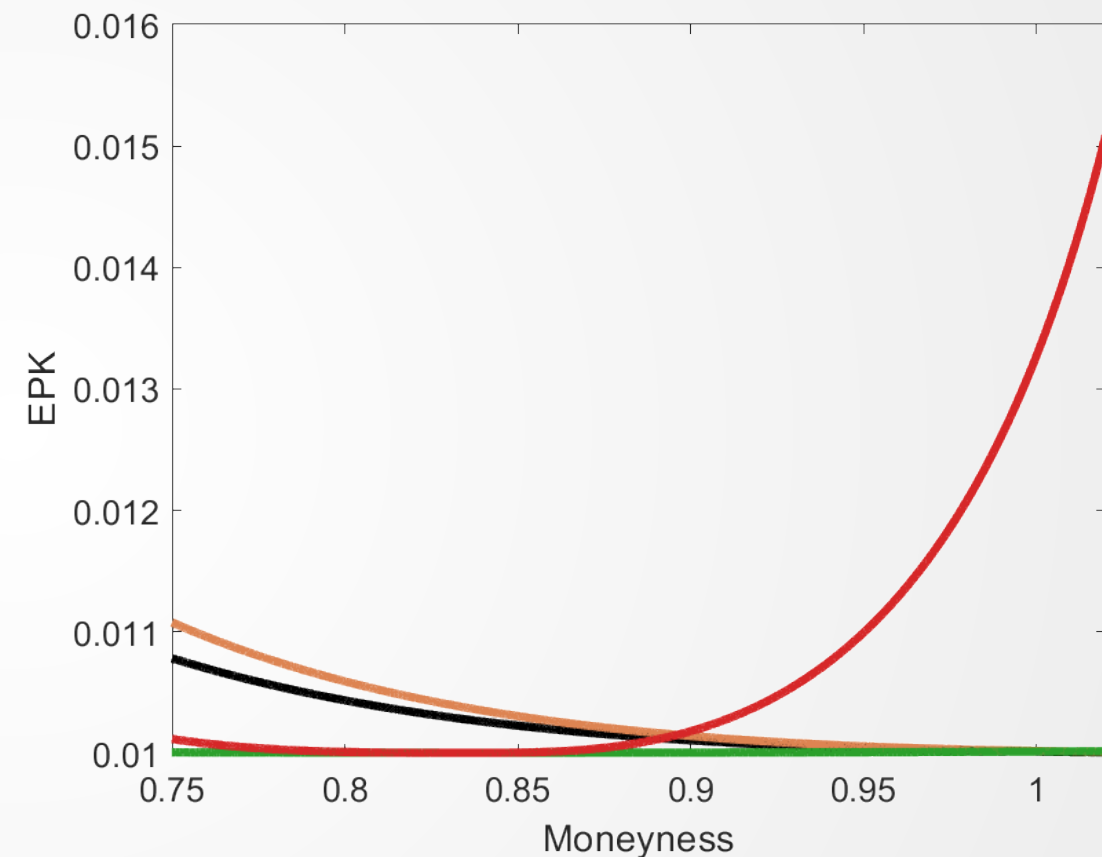
- Monotonic decreasing does not always hold in this market



CDI result of of Bitcoin Option

2022-11-17 $\tau = 8$

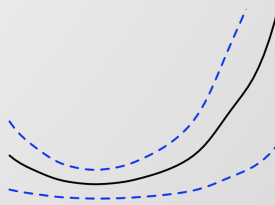
4 moments, 4 knots; 5 moments, 5 knots;
6 moments, 6 knots; 7 moments, 7 knots



CDI result of of Bitcoin Option

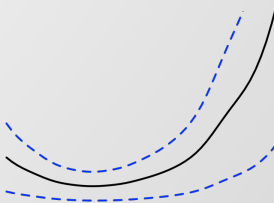
2022-10-29 $\tau = 13$

4 moments, 4 knots; 5 moments, 5 knots;
6 moments, 6 knots; 7 moments, 7 knots

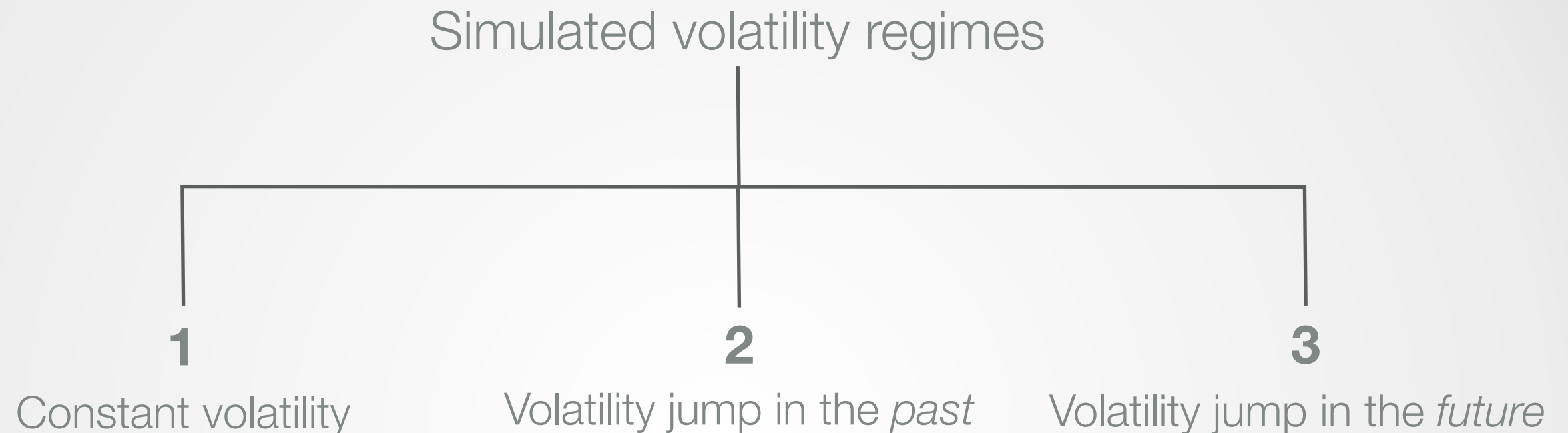


Potential reason of non-monotonicity

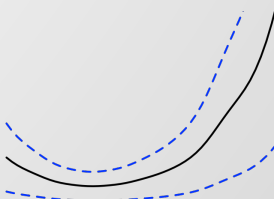
- ▣ Volatility is not constant in different trading dates
- ▣ Sensitive to the choice of moments and knots parameters in CDI
- ▣ To avoid inconsistent methodology problems, kernel estimation should replace B-spline



It's the vola, stupid!



- ▣ Analyze the impact of volatility regime switches on the two proposed PK estimations, we examine the PKs at the same point in time under different overall volatility regime scenarios.
- ▣ The classical PK estimation relies on historical returns. It is to be assumed that a volatility jump in the past influences the kernel, while a volatility jump in the future does not.
- ▣ In contrast, the CDI PK estimation uses forward-looking returns. In theory past regime switches have no effect on the pricing kernel compared to future regime switches.



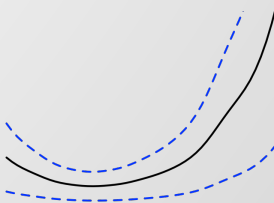
Case 1: Constant volatility



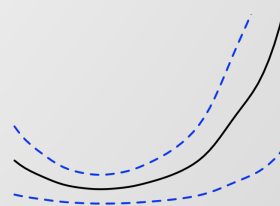
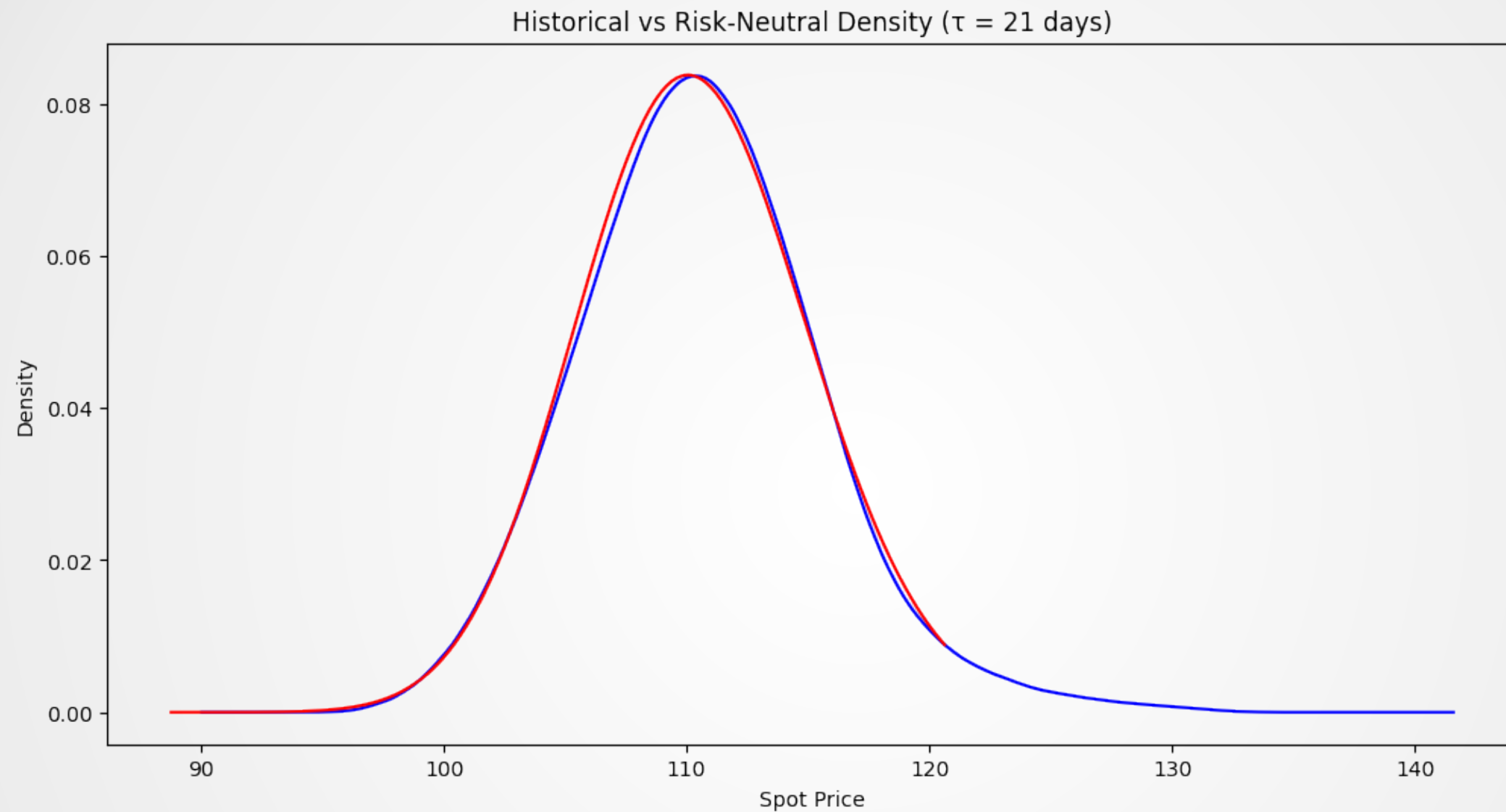
Price and Volatility Path



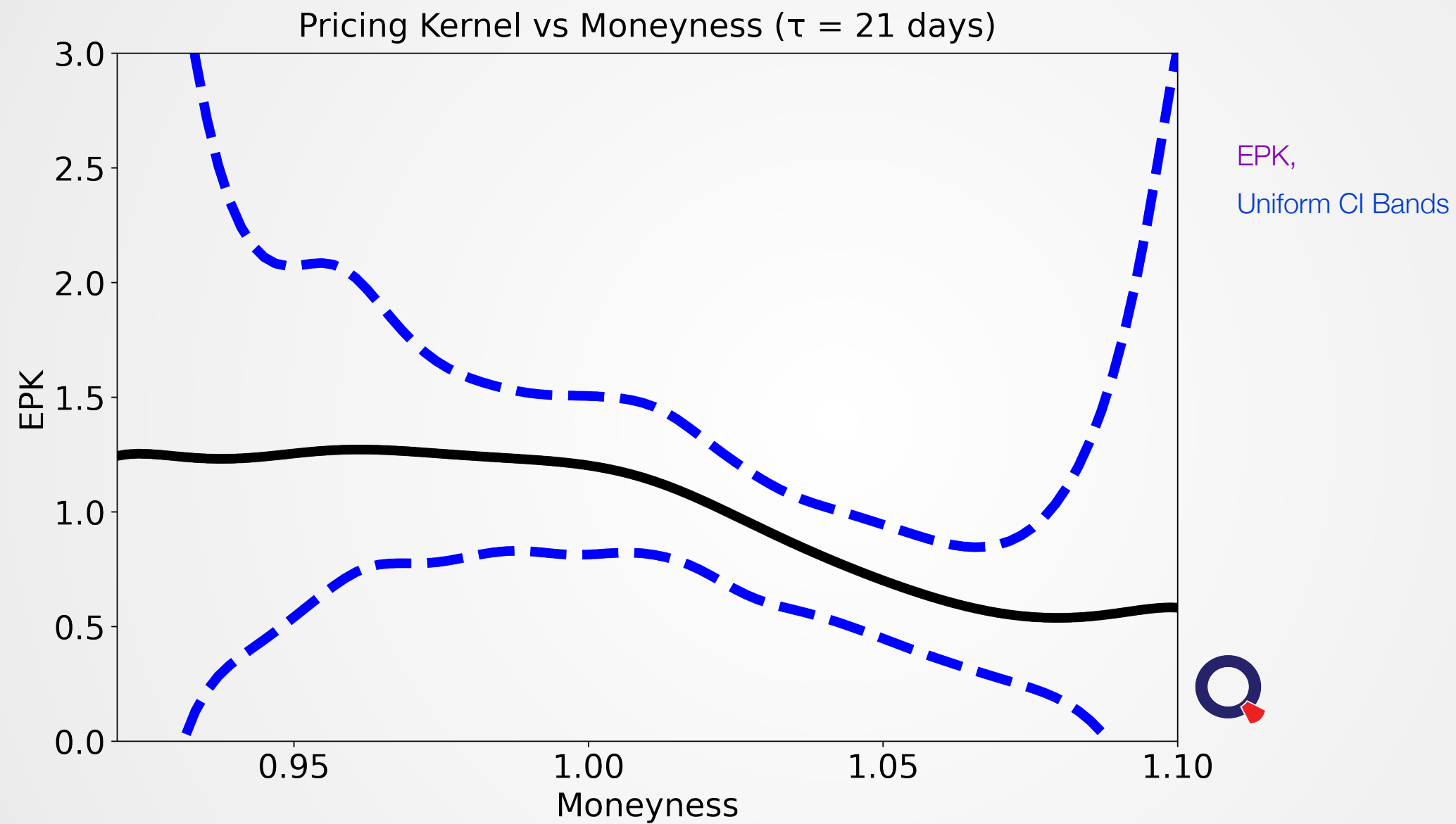
□ Estimation of the PK



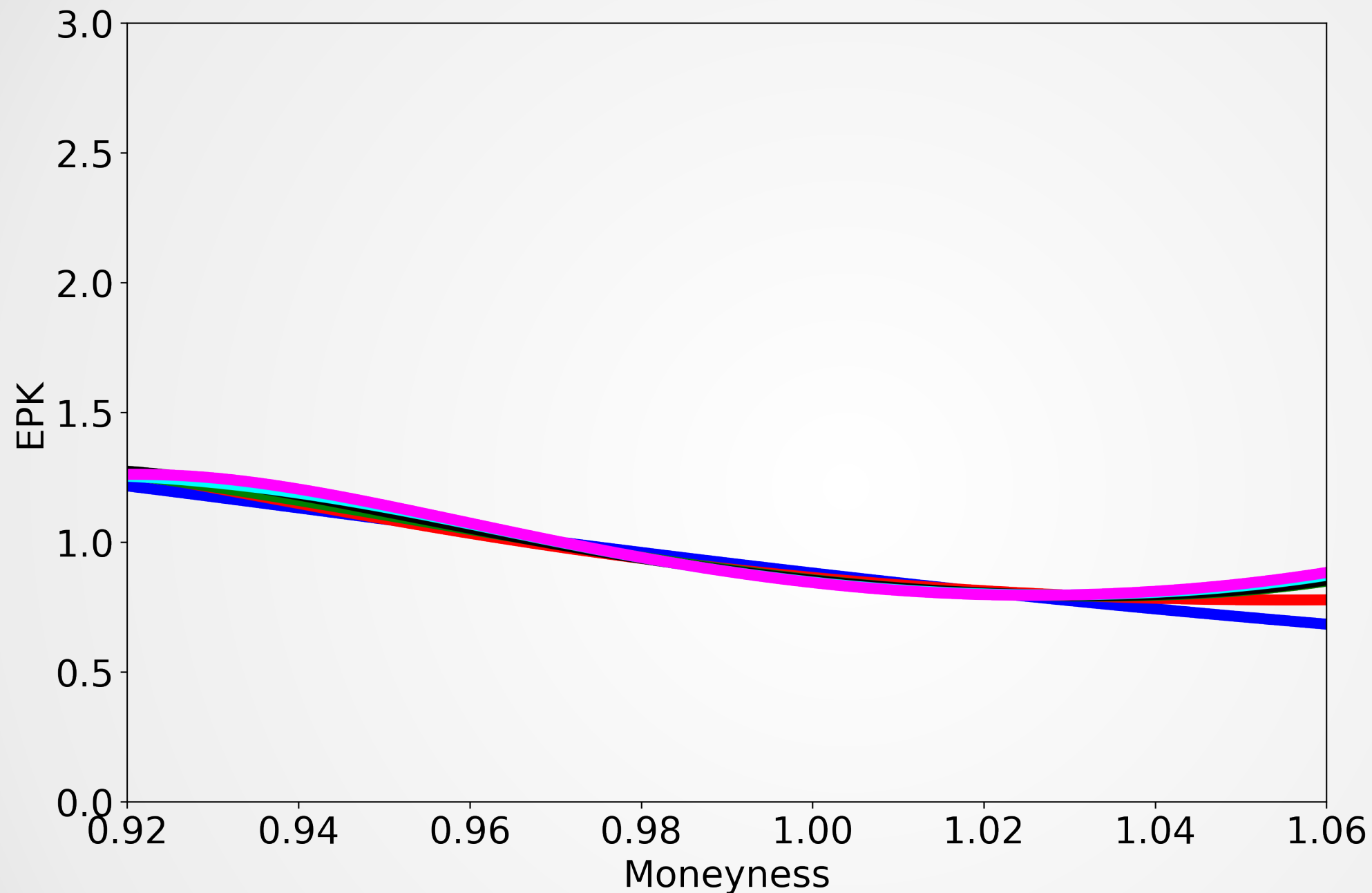
Case 1: RND and HD at constant volatility



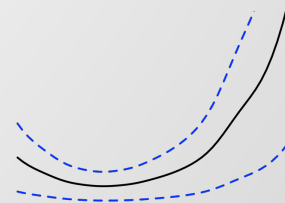
Case 1: Classical Pricing Kernel



Case 1: CDI Pricing Kernel of constant volatility



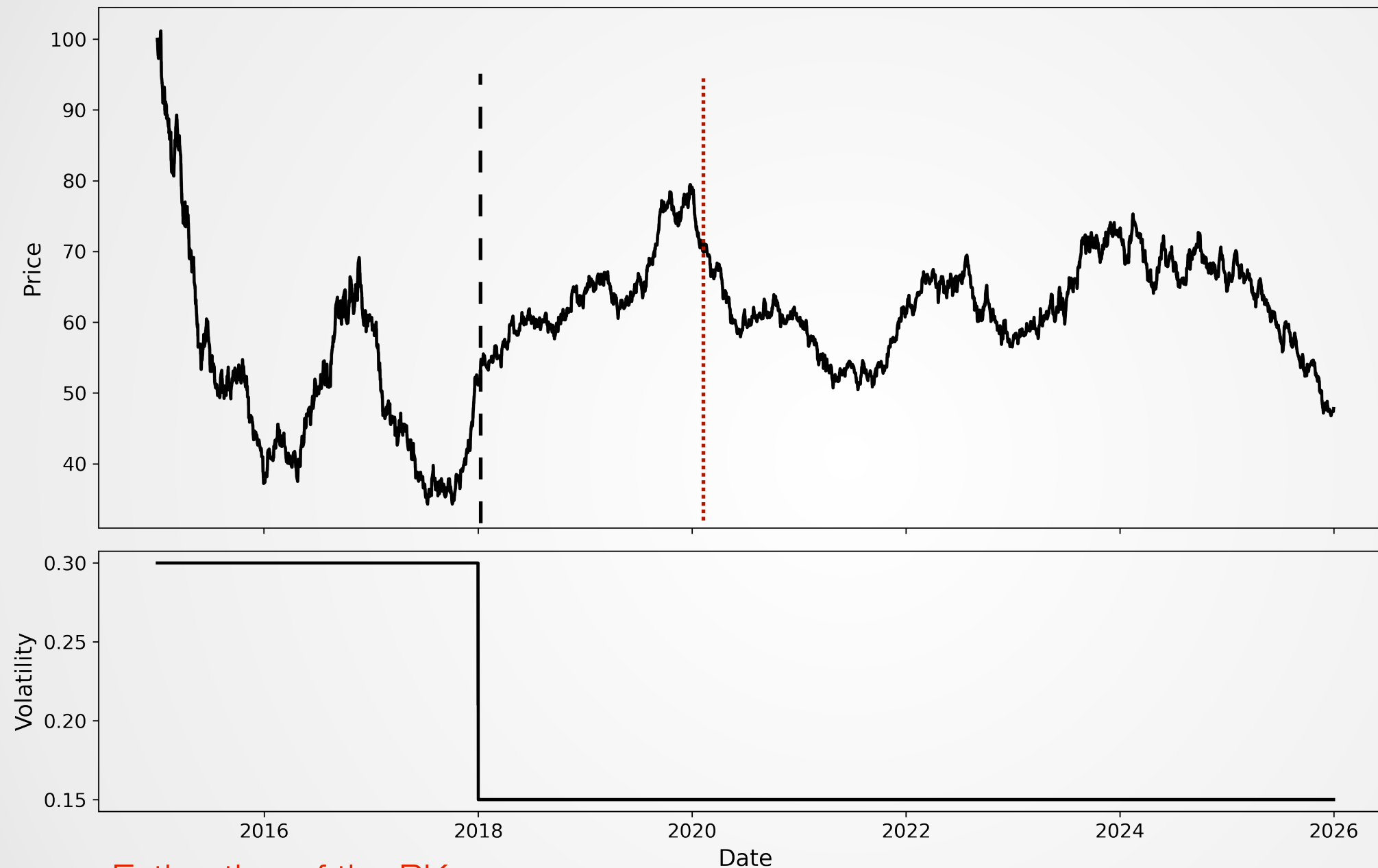
4 bases/moments, 5 bases/moments,
6 bases/moments, 7 bases/moments,
8 bases/moments, 9 bases/moments
Steps: 0.001



Case 2: Volatility jump in the *past*

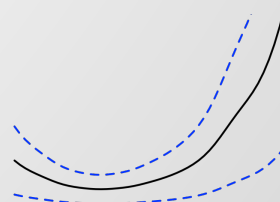


Price and Volatility Path



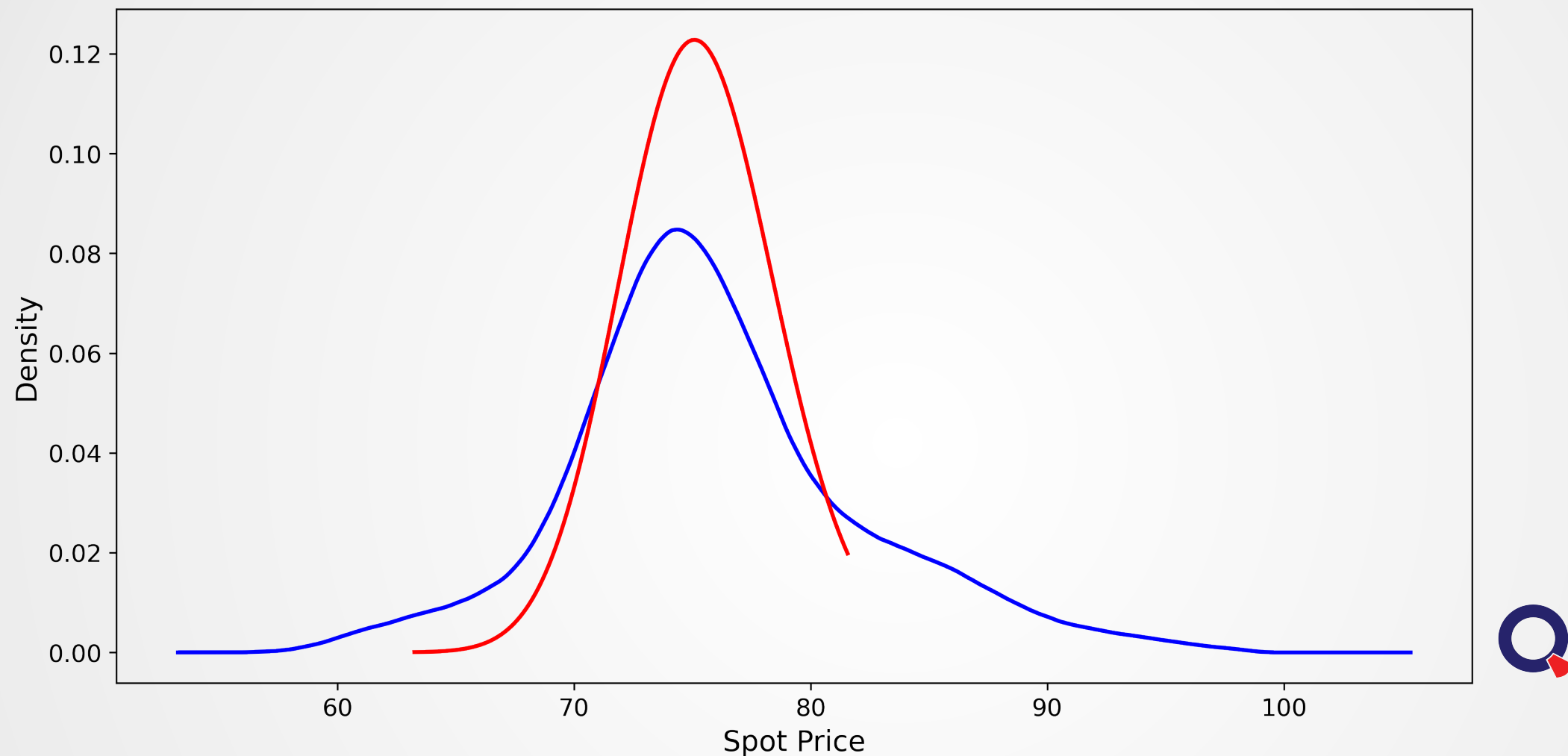
▣ Estimation of the PK,

▣ Volatility jump: $\sigma(t) = \begin{cases} 0.3 & \text{for } t < 2018, \\ 0.15 & \text{for } t \geq 2018. \end{cases}$



Case 2: RND and HD

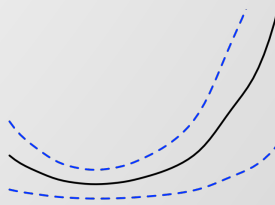
Historical vs Risk-Neutral Density
 $\tau = 21$ days



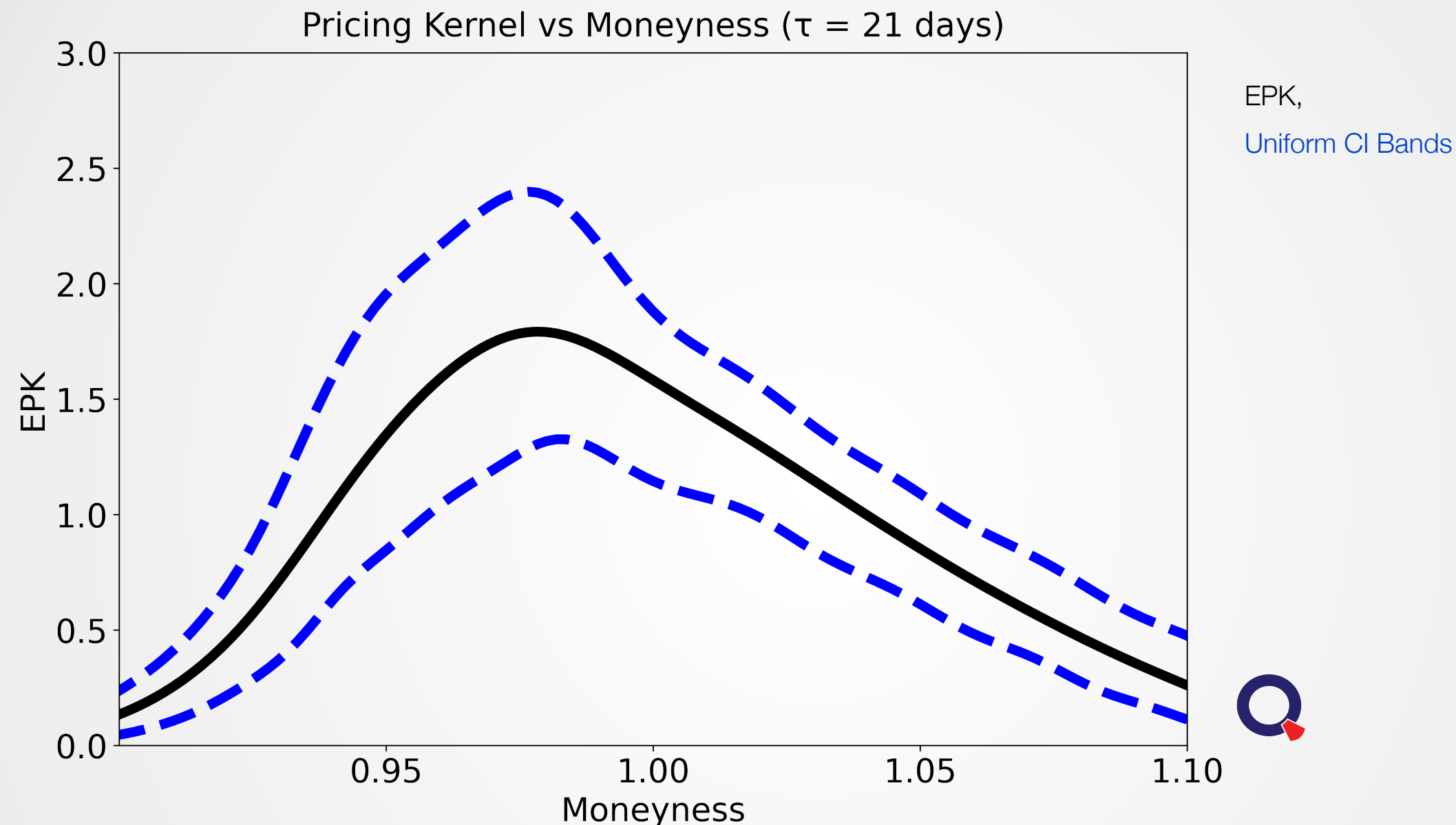
▣ RND and HD at 2020-01-10 with

▶ $r = 0.00$

▶ $\tau = 21$

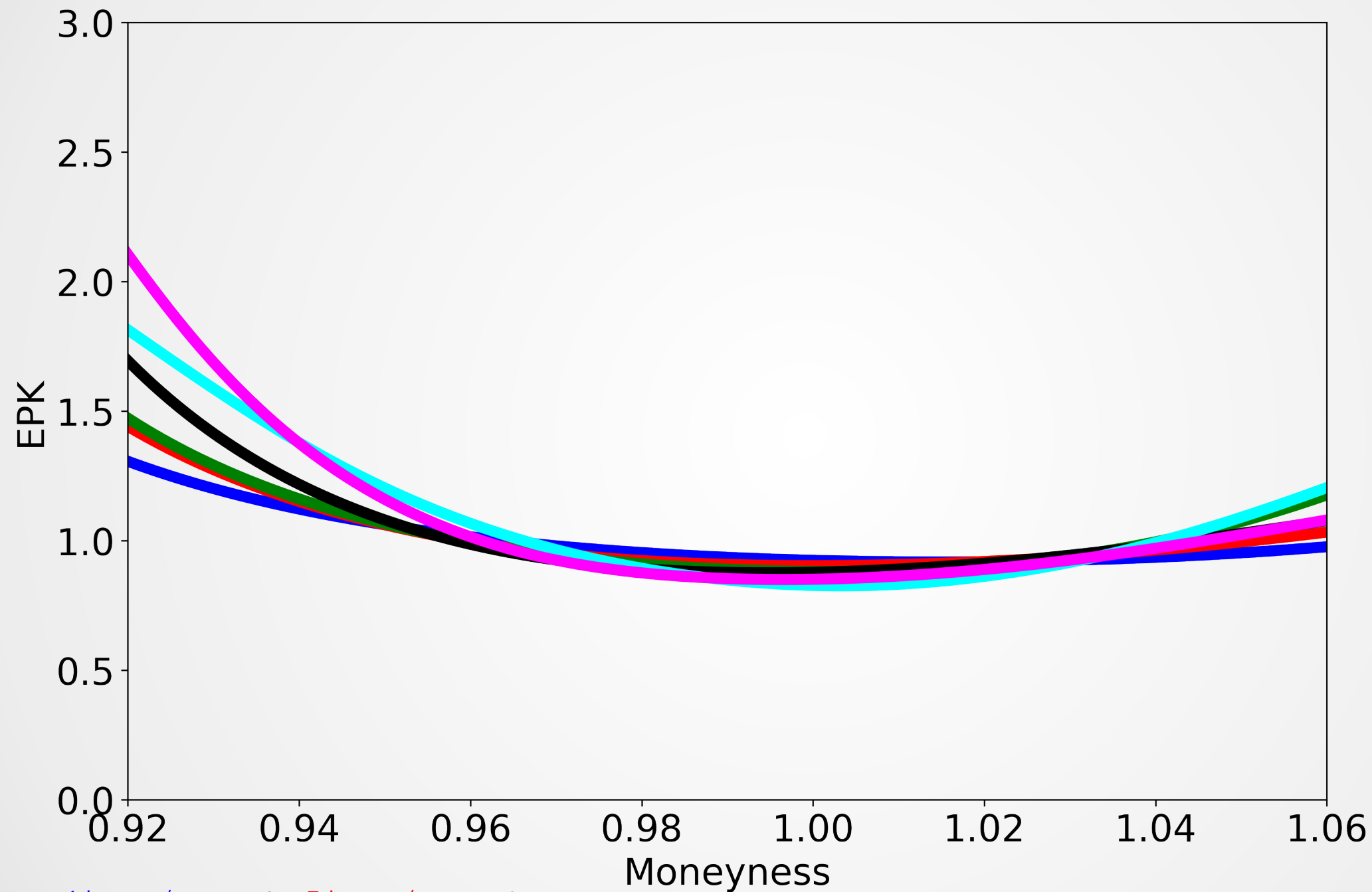


Case 2: Classical PK

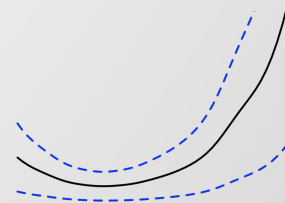


- The historical density estimation is based on two distinct volatility regimes. This mixed-regime historical density significantly influences the kernel shape.

Case 2: CDI Pricing Kernel of Jump in the Past

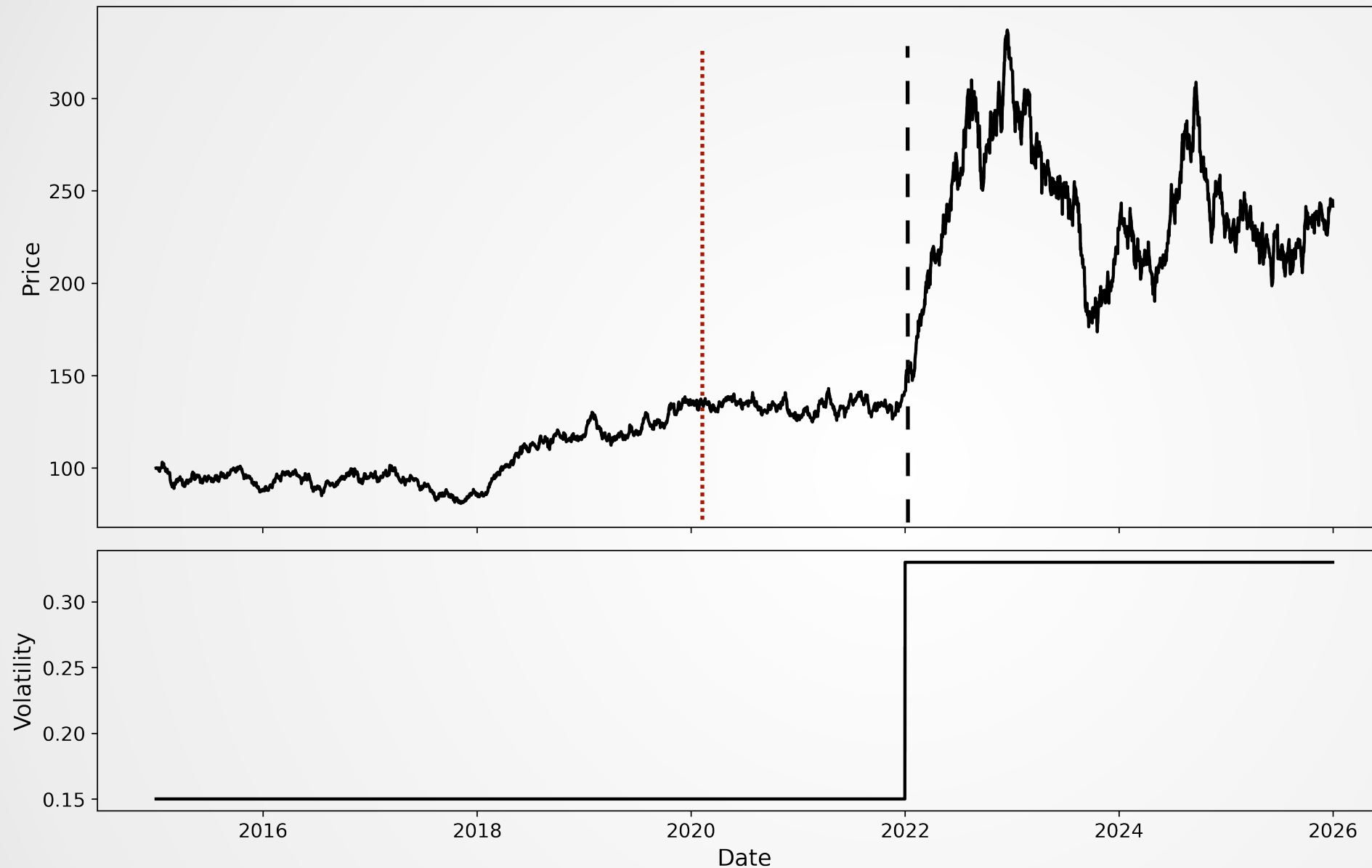


4 bases/moments, 5 bases/moments,
6 bases/moments, 7 bases/moments,
8 bases/moments, 9 bases/moments
Steps: 0.001



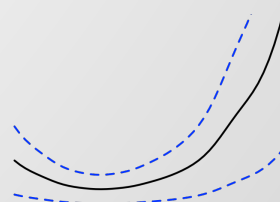
Case 3: Volatility jump in the "future"

Price and Volatility Path

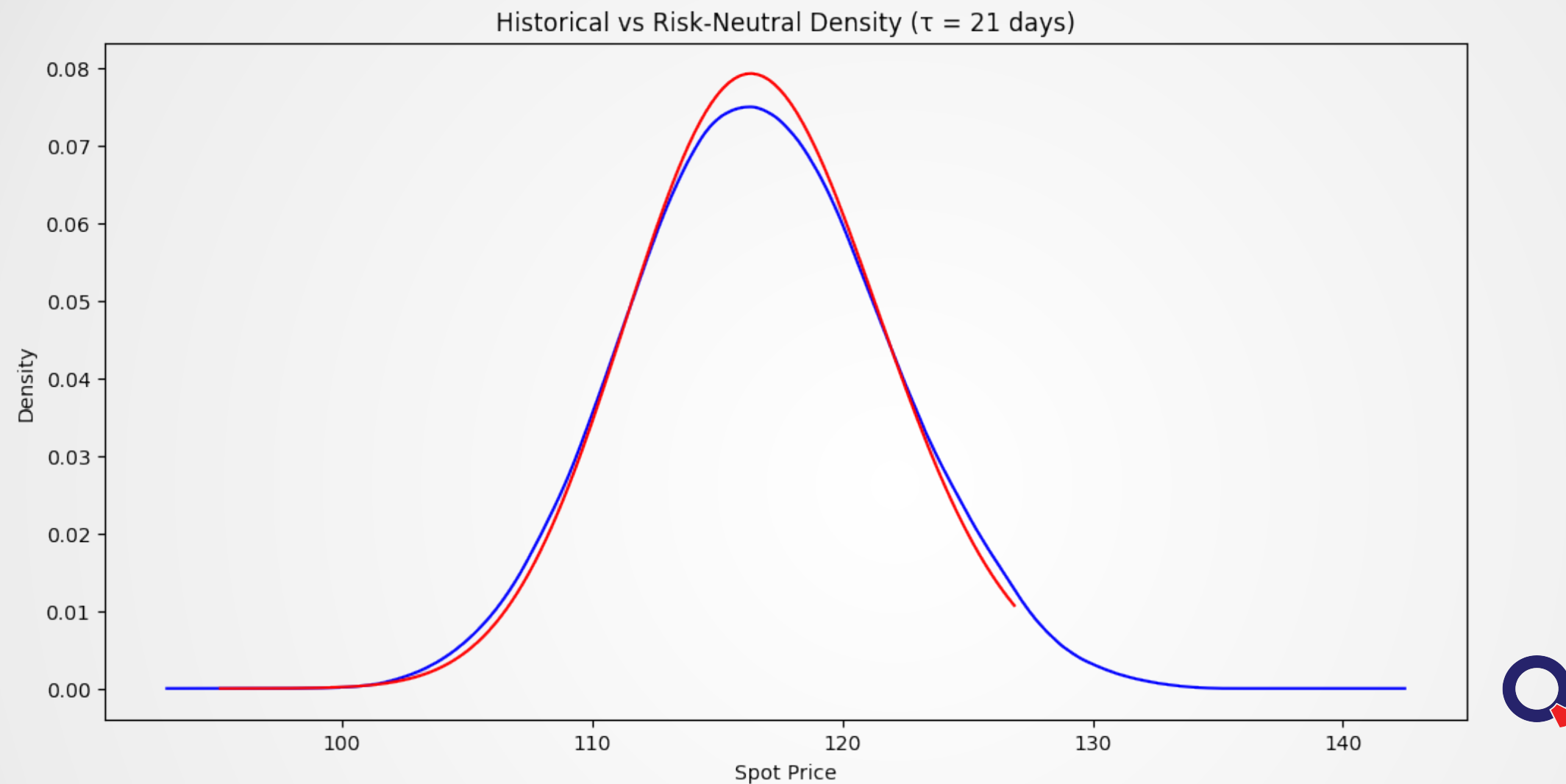


▣ Estimation of the PK,

▣ Volatility jump: $\sigma(t) = \begin{cases} 0.15 & \text{for } t < 2022, \\ 0.3 & \text{for } t \geq 2022. \end{cases}$



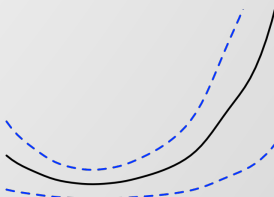
Case 3: RND and HD



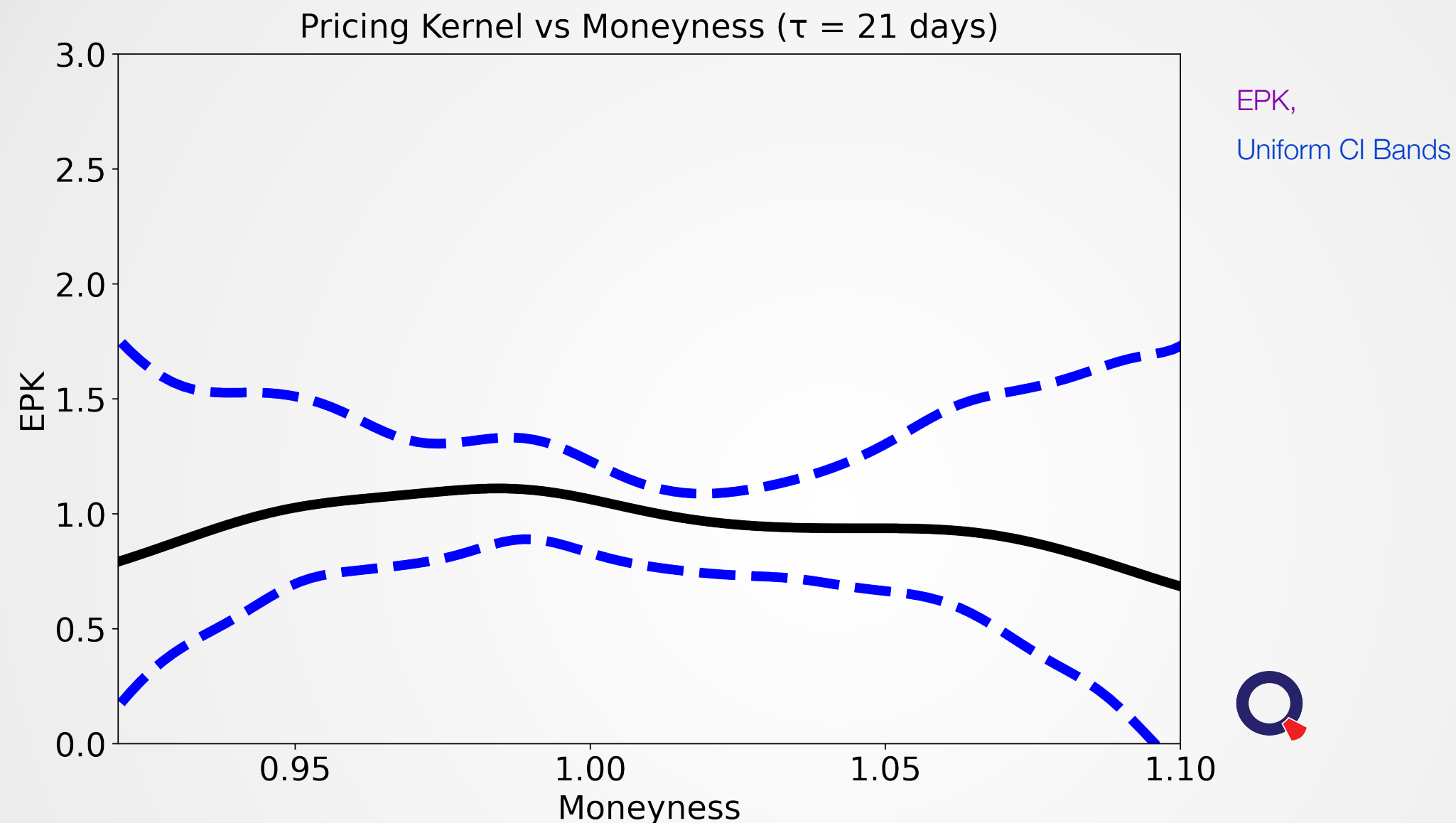
▣ RND and HD at 2020-01-10 with

▶ $r = 0.00$

▶ $\tau = 21$

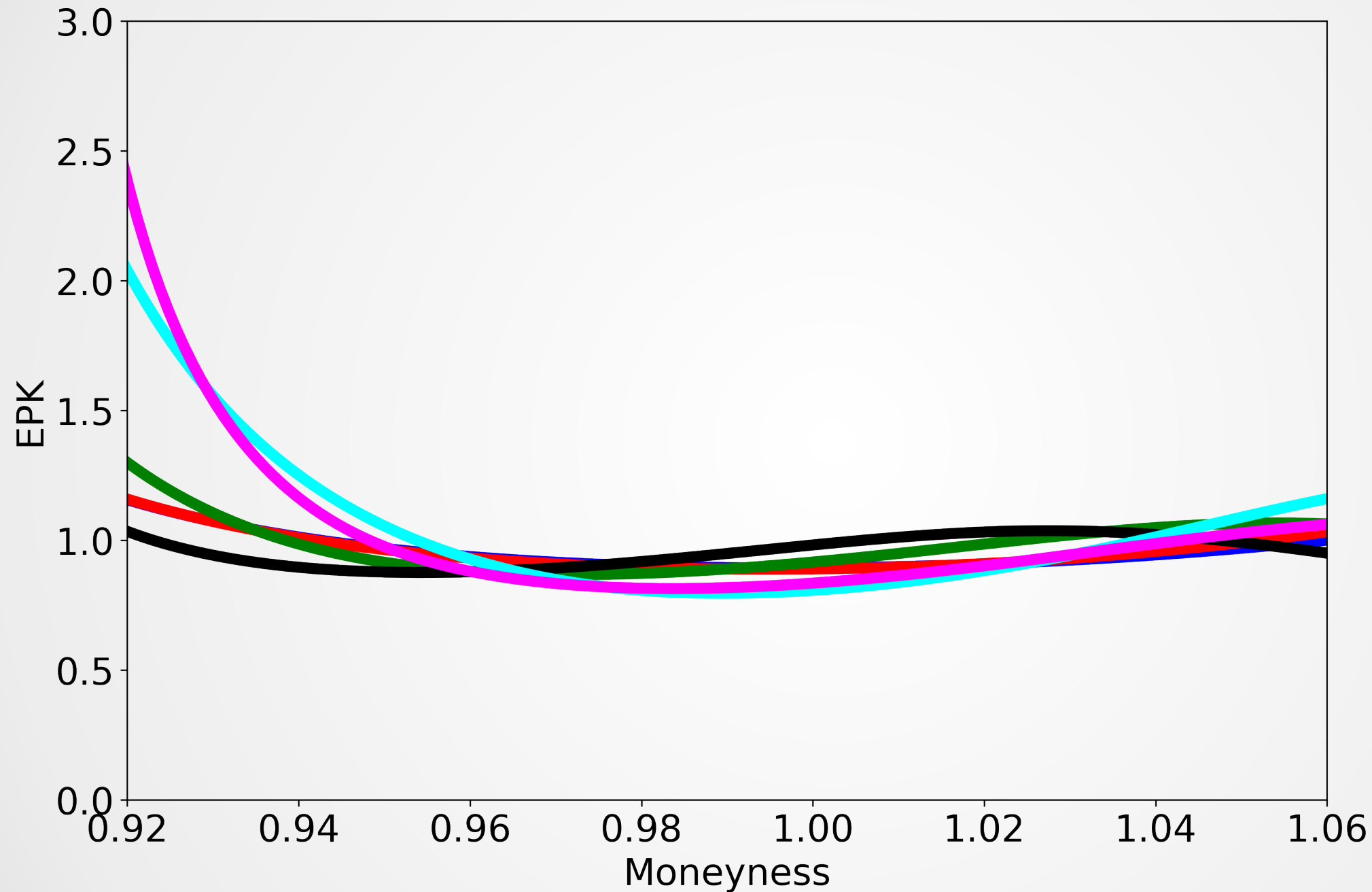


Case 3: Classical Pricing Kernel

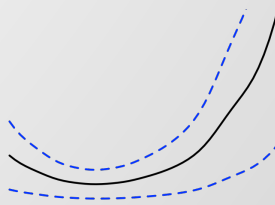


- A volatility jump in the *future* has no effect on the characteristics of the PK, as only *past* data is being used for the estimation.
- The PK is similar to the baseline kernel.

Case 3: CDI Pricing Kernel of Jump in the Future

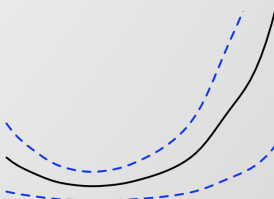


4 bases/moments, 5 bases/moments,
6 bases/moments, 7 bases/moments,
8 bases/moments, 9 bases/moments
Steps: 0.001



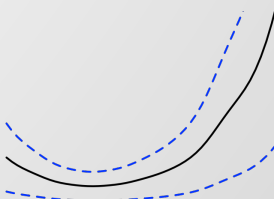
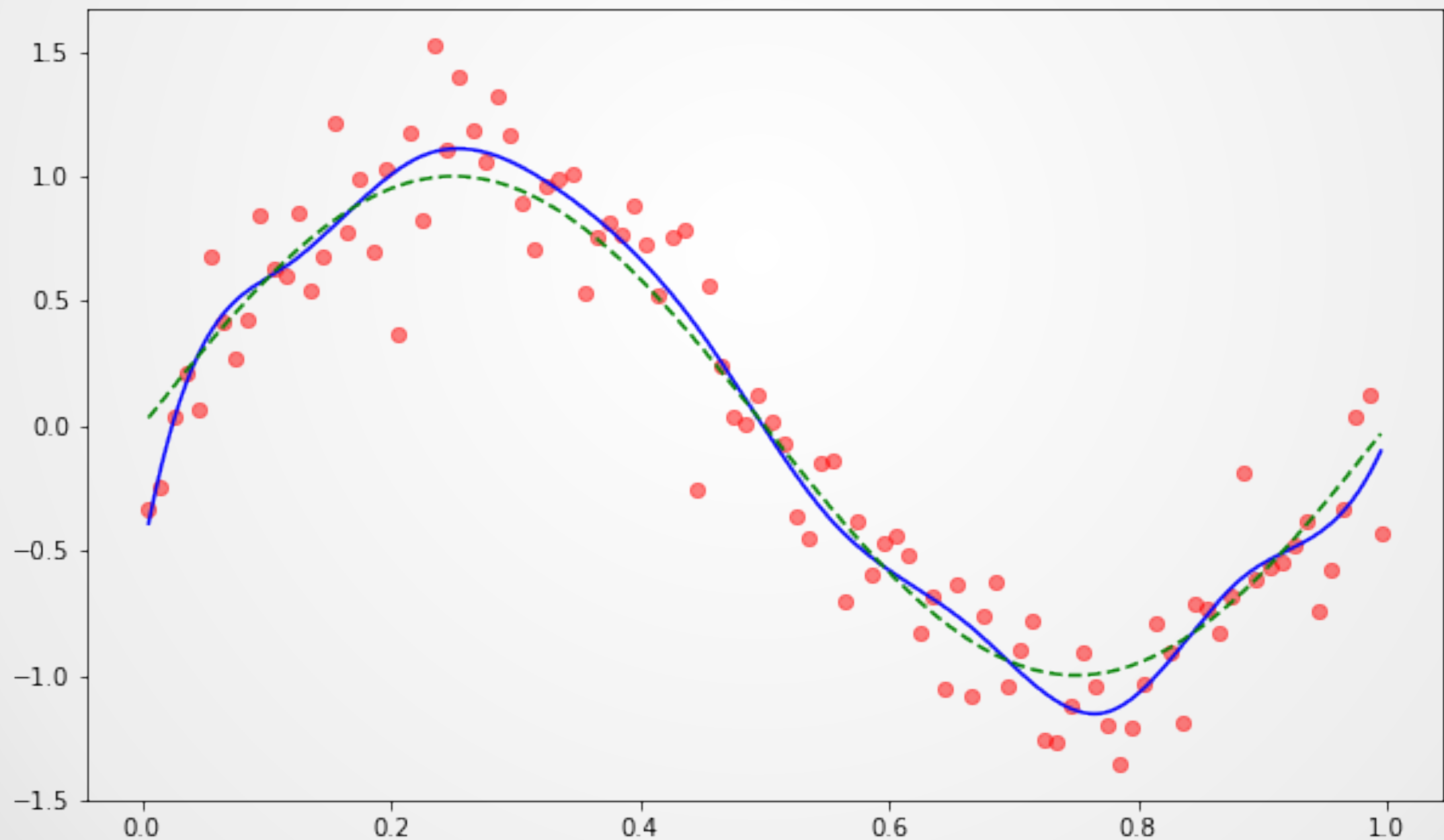
Highlights

- ▣ Previous literature compares mismatched information sets
- ▣ The CDI approach is a nonparametric estimation technique using B-Splines for the pricing kernel by using GMM minimization
 - ▶ Requires daily returns and risk-neutral density as input
 - ▶ It includes forward looking returns as upper bound within the integral
 - ▶ Asymptotic distribution theory of Empirical Pricing Kernel (EPK) is lacking
 - ▶ Test for (non)monotone shape is impossible. Alternatively:
Uniform confidence band of EPK
- ▣ CDI averages a period of time
- ▣ # knots in the B-spline should be carefully selected



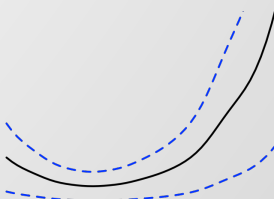
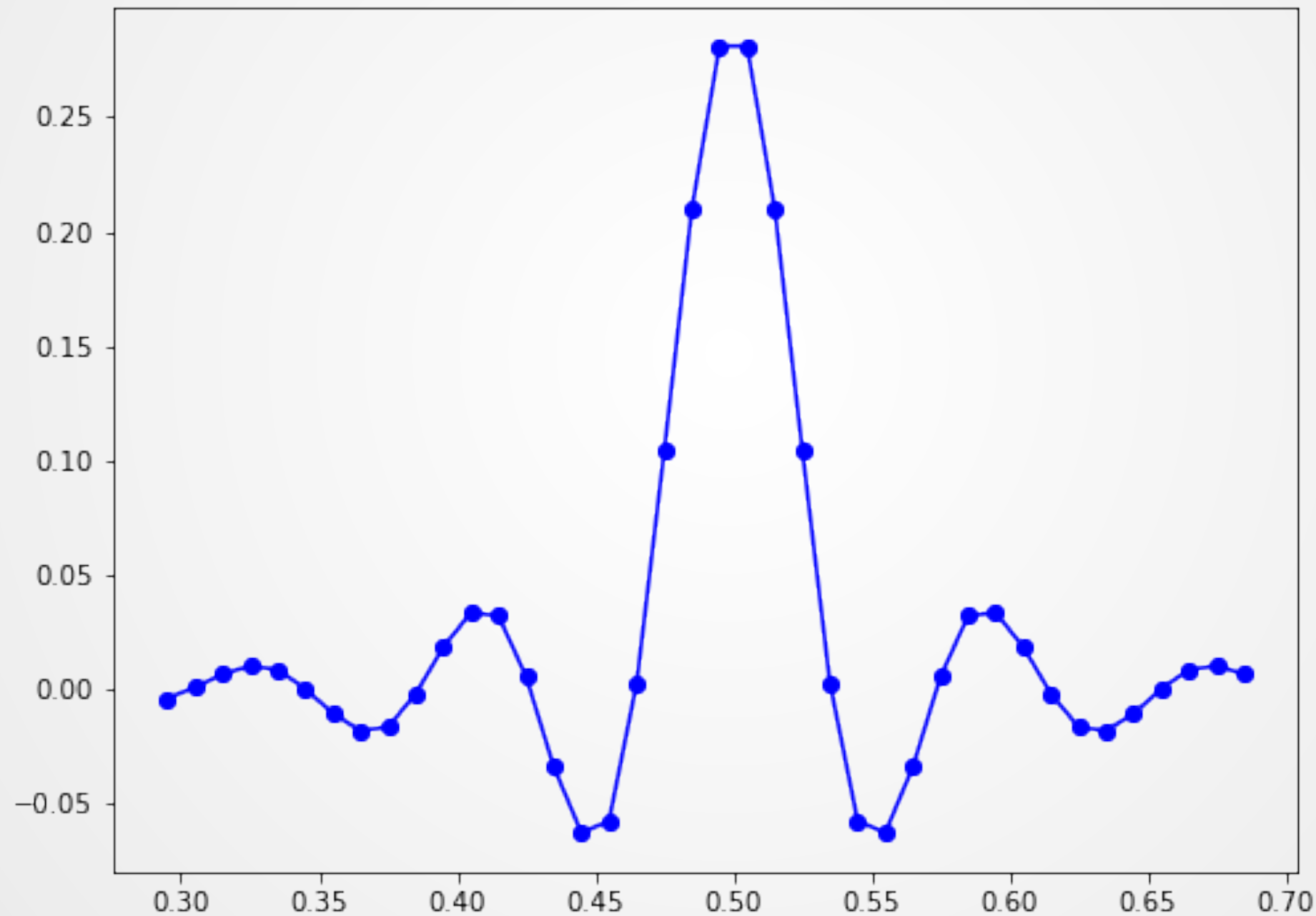
omnes viae Romam ducunt

- B-spline (deg=3, k=10) smooth on $\sin(2\pi x)$ with noise (sig=0.2)



All smoothers are local averages

- B-spline contributions at $x = 0.50$

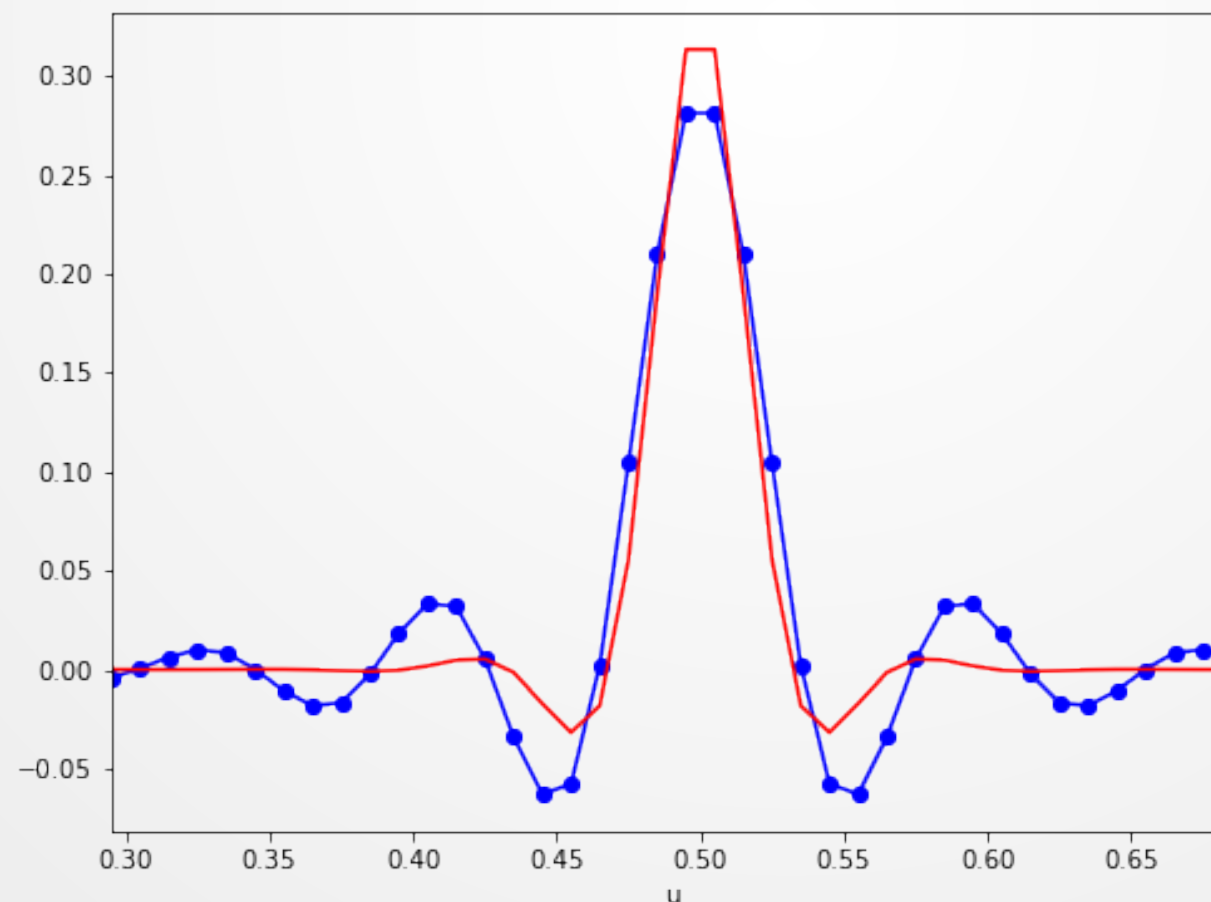


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$$f(u) = \frac{1}{6} \cdot \exp(|u|) + 2 \cdot \exp\left(-\frac{|u|}{2}\right) \cdot \sin\left(\frac{\pi}{6} + \sqrt{3} \cdot \frac{|u|}{2}\right)$$

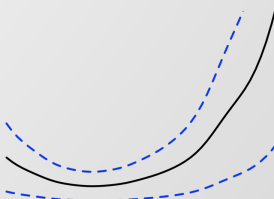
$$G(x, x_i) = f\{(x - x_i)/h\}/h; h - \text{bandwidth}$$

$$\hat{m}(x) = n^{-1} \sum G(x, x_i) \cdot Y_i$$



$f(u)$

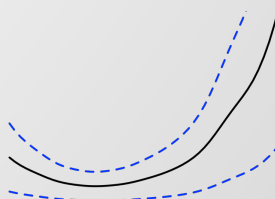
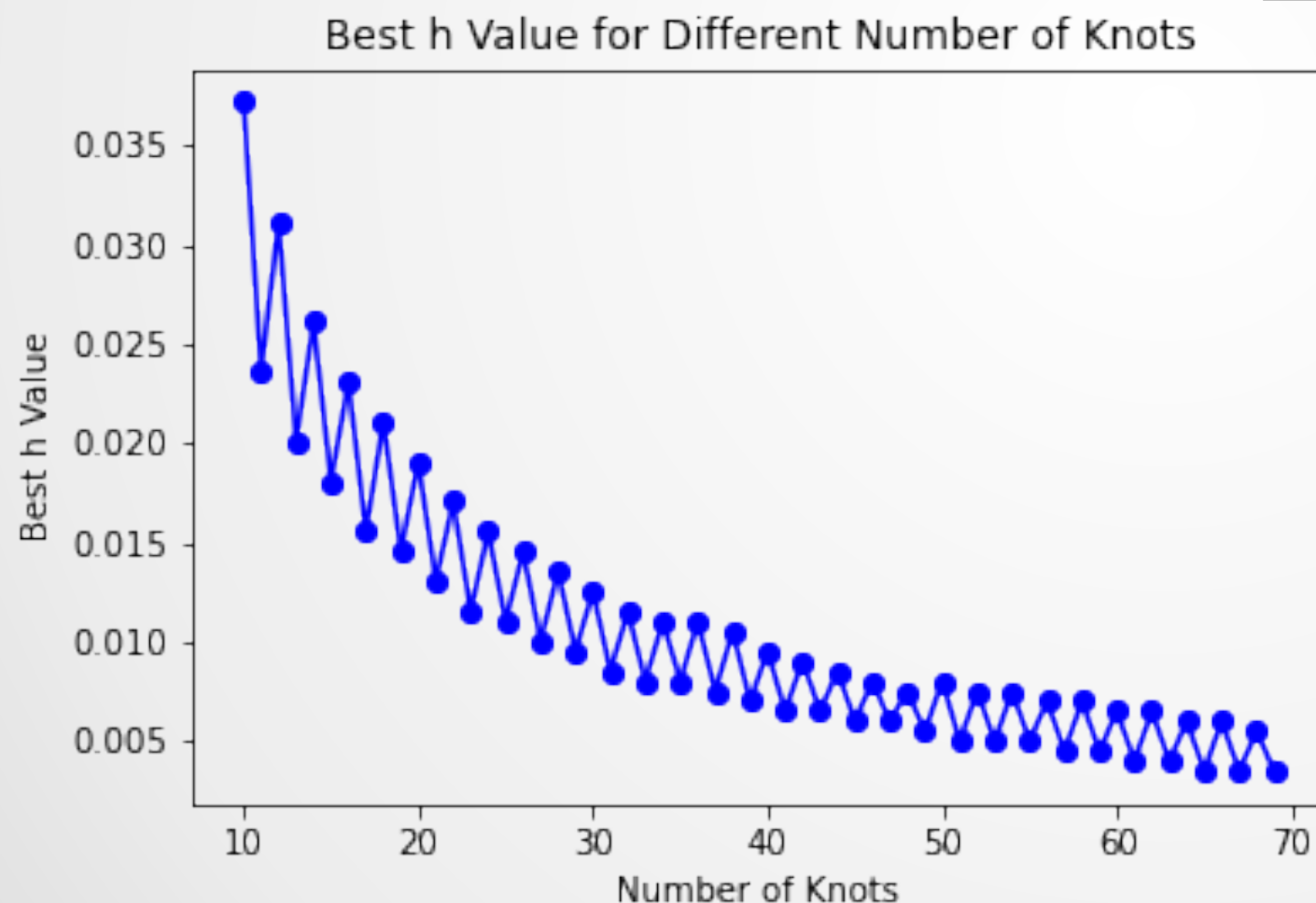
B-spline



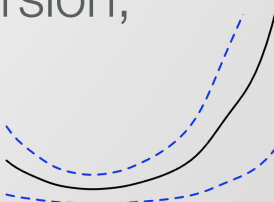
#knots = 1/h

- Given k , find the best h , to minimize the SSE between two smoothed curves
- $\log(k) = \beta_0 + \beta_1 \log(h) + \varepsilon$

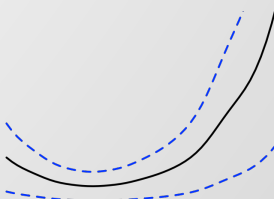
Y	log(h)
log(k)	-1.011*** (0.047)
Observations	60
Adj_R^2	0.885



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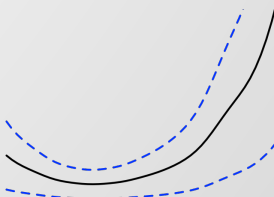


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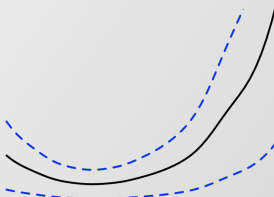
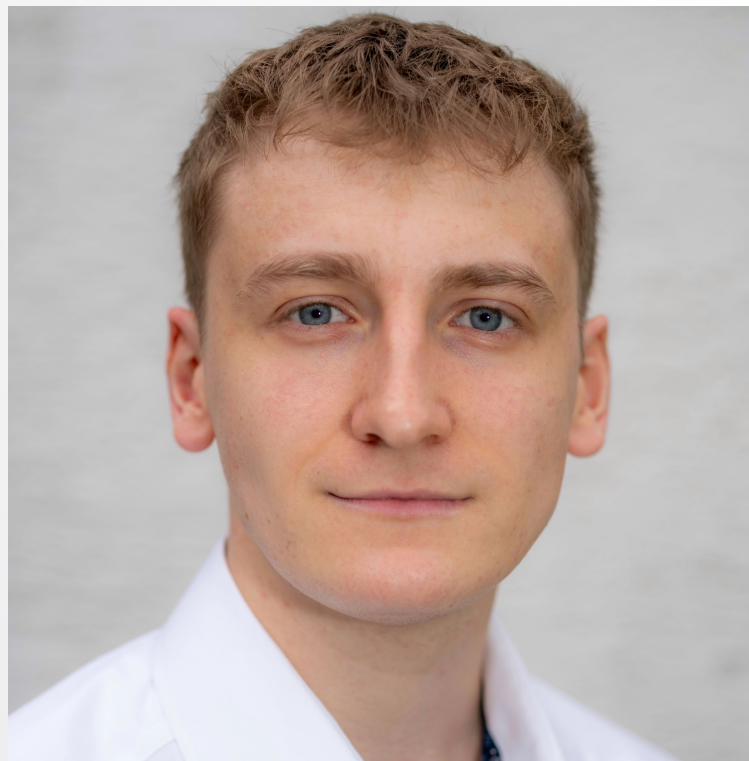


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thanks to Yuri Golubev and Ratmir Miftachov and Yanchu LIU



Team



Management & Marketing

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Call for Papers

Generative AI in Finance: Innovation, Risk, and Systemic Stability



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This special collection focuses on the rapidly growing role of **generative artificial intelligence (AI)** in transforming financial markets, institutions, and decision-making processes. Recent advances in large language models (LLM), generative neural networks, and foundation models have enabled new forms of data generation, prediction, and automation that extend well beyond traditional machine learning approaches. These developments are reshaping financial modeling, risk management, trading, regulatory compliance, and customer-facing financial services.

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- Case studies and real-world implementations
- Macroeconomic and financial stability implications of generative AI
- Future directions and emerging trends

This collection seeks to foster interdisciplinary dialogue among researchers in finance, economics, computer science, and policy, and to advance both academic understanding and practical insights into the evolving role of generative AI in the financial ecosystem.

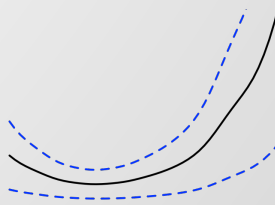
Guest Editors

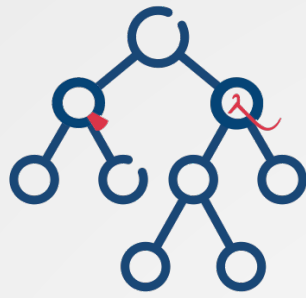
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Ruting Wang, City University of Hong Kong, Hong Kong
Xinheng Liu, Changsha University of Science and Technology, China
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Submission Deadline: **31 December 2026**



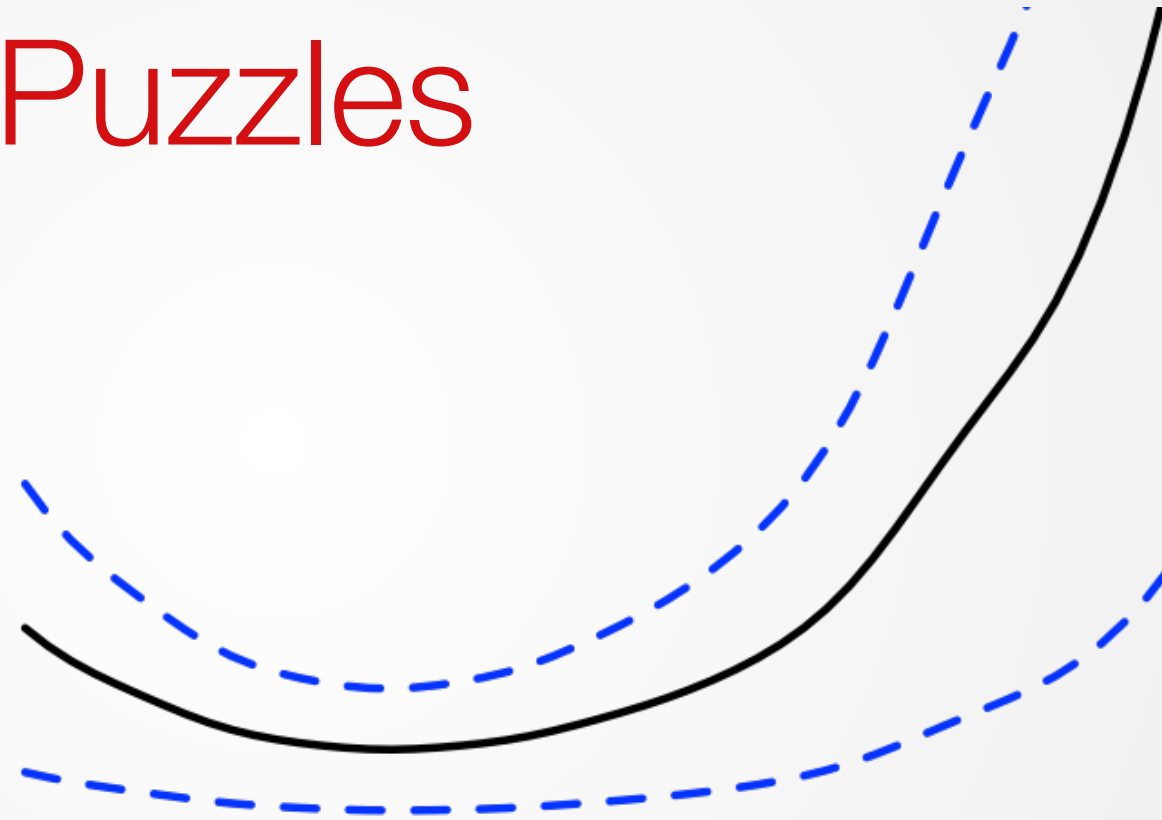


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Pricing Kernel Puzzles

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